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1.0 Project Initiation

1.1 Project Overview

EasyAppointment is a healthcare and appointment scheduling system designed to be offered by the Jordanian Ministry of Health to navigate and search for the right doctor in any medical facility, both Governmental and private. It aims to make it easy for all citizens to find the best doctor for them, considering their health condition, location, and financial capacity. With EasyAppointment, every citizen will have a medical ID, allowing doctors to access all necessary medical information before seeing the patient. This system will facilitate better healthcare outcomes by ensuring that medical professionals have comprehensive patient histories at their fingertips.

1.2 Problem Definition

1.2.1 Problem Statement

Jordanian citizens face significant difficulties when dealing with government transactions, especially in the medical sector. These challenges include obtaining necessary treatment due to a lack of information and the complexity of the current system.

While private hospitals have made some improvements, they are often unaffordable for many citizens and may not have specialized doctors for complex conditions. Consequently, there remain numerous problems with the existing healthcare situation.

1.2.2 Issues

A. Lack of Information: Government healthcare services provide insufficient information, making it difficult for citizens to navigate the system and find appropriate medical care.

B. Complex System: The current process of accessing medical services is overly complex, leading to frustration and delays in receiving treatment.

C. Limited Affordability: Private hospitals, although more streamlined, are often unaffordable for a significant portion of the population.

D. Specialized Care Deficiency: Not all hospitals, including private ones, have the necessary specialized doctors for complex

medical conditions and that makes it hard to find the needed doctor.

1.2.3 Objectives for each issue

☐ **Lack of Information:**

Create a comprehensive, user-friendly information system to help citizens easily access necessary details about healthcare services and providers.

☐ **Complex System:**

Simplify the process of finding and scheduling medical appointments, reducing bureaucracy and streamlining access to care.

☐ **Limited Affordability:**

Develop a system that allows patients to easily navigate and search for affordable healthcare options. By integrating public and private healthcare resources, the system will provide a comprehensive range of healthcare prices, ensuring accessibility for all financial capacities.

☐ **Specialized Care Deficiency:**

By integrating all healthcare providers, it will make it easy to find a Specialized doctor for each medical case.

1.2.4 Requirements

Table 1: table of requirements

Requirement	Description
User-Friendly Interface	The platform must be intuitive and user-friendly, ensuring accessibility for all citizens regardless of their technical expertise.
Comprehensive Information Database	The system must include a detailed database of healthcare providers, listing qualifications, specializations, locations, and available services.
Medical ID System	Each user should have a unique medical ID that consolidates their medical history, making it accessible to healthcare providers.
Simplified Appointment Scheduling	The system must feature an intuitive scheduling tool for booking and managing appointments in real-time.
User Accounts and Roles	Each user must have a secure, personal account to store personal information, medical history and appointment schedules, also it must support different user roles.
Secure Data Management	The system database holds Sensitive data so it must ensure the privacy and security of user data.

Appointment Reminders	Send automated reminders for upcoming appointments to patients and doctors.
Specialized Care Identification	The system should provide tools to help users find specialized doctors and facilities suitable for their specific health conditions.

1.2.5 Constraints

The EasyAppointment project is subject to several constraints that may impact its implementation and success:

- A. Society and People Adoption:** The project must consider the varying levels of technical expertise among users, particularly those with low technical skills. Ensuring ease of use and providing adequate support will be crucial for widespread adoption.
- B. Private Sector Cooperation:** Successful implementation will require collaboration with private healthcare providers. Securing their cooperation and integrating their services into the system may present challenges.

- C. **Data Collection for Initial Setup:** Collecting comprehensive and accurate data from both public and private healthcare providers during the initial phases of the project will be a significant challenge. Establishing data-sharing agreements and ensuring data accuracy are critical.

- D. **Financial Support:** Adequate funding is essential for the development, implementation, and maintenance of the system. Securing financial support from the government, private sector, and possibly international organizations will be necessary.

- E. **Marketing and Awareness:** The project's success depends on public awareness and acceptance. Effective marketing strategies will be needed to inform citizens about the system and encourage its use.

1.3 Feasibility Study

1.3.1 Technical Feasibility

A. Technological Capabilities: Considering the advancements in app development techniques. It is very feasible to develop an app or website like “EasyAppointment” using Swift, Kotlin for apps, or React, Angular for websites and firebase for security.

B. Data Integration: Integration with existing healthcare systems may present challenges, especially if these systems are outdated or use varying data standards. However, with the expertise of our staff, and using cloud-based database services for implementing and integrating databases, it is achievable.

C. User Interface: Designing a user-friendly interface is very feasible, given the availability of advanced UI/UX design practices and tools.

1.3.2 Operational Feasibility

A. User Adoption: The platform is designed to be intuitive and user-friendly, making it accessible to citizens of varying technical expertise. This ensures ease of use and encourages widespread adoption.

B. Integration with Existing Infrastructure: By leveraging existing public and private healthcare resources, EasyAppointment can provide a comprehensive database of healthcare providers, ensuring users have access to a wide range of services.

C. Endorsement by the Ministry of Health: The system will be offered by the Jordanian Ministry of Health, providing it with credibility and ensuring that it meets national healthcare standards. This endorsement will also aid in securing cooperation from public and private healthcare providers.

D. Resource Availability: The availability of skilled personnel wouldn't be a problem, due to Jordan having a plethora of skilled programmers and IT technicians. However, due to the large databases to be used in the app. Using physical servers would prove costly. Therefore, cloud-based databases will be used to accommodate the large databases used.

1.3.3 Financial Feasibility

Table 2: financial feasibility table

Category	Description	Quantity	Unit Price (JD)	Total Price (JD)	Notes
Hardware Requirements					
Developers Workstations	High performance laptops	3	950	2850	For app development and design tasks
Cloud Services					

Virtual Machines to host the website	Cloud-based Azure hosting service	1	150 (per month)	12000 (per year)	To host the “EasyAppointment” website
Relational Database Service	Microsoft Azure’s cloud Database (16 cores, 128 GB RAM)	1	1000 (per month)	1800 (per year)	For storing, managing, and querying user and health data
Object Storage	Cloud based (Azure Blob Storage)	1	200 (per month)	2400 (per year)	For storing files, images, and patient records securely

Employee costs and salaries					
System architect	Ensures all parts of the system are working together seamlessly	1	1500 (per month)	18000 (per year)	
UI/UX designer	Ensures that the app is user-friendly, aesthetically pleasing, and easy to navigate		800 (per month)	9600 (per year)	

Full stack developer	Responsible for both front-end developer and back- end developmen t	1	1200 (per month)	14400 (per year)	
Database administrator	Responsible for managing the database.	1	900 (per month)	10800 (per year)	
Total Yearly Costs (JD)	2850				

Total Yearly	52800
Costs (JD)	

1.3.4 Schedule Feasibility

Table 3: schedule feasibility table

Task ID	Task Description	Duration	Dependencies	Responsible staff	resources
T1	Initial Stake Holder meeting	5	None	All team member	None
T2	Requirements specifications	7	T1	Ayham (BA)	None
T3	System Architecture design	14	T2	Yazan (System Architect)	Lucid chart
T4	Detailed system design	12	T3	Yazan, Ibrahim (system design team)	starUML
T5	UI/UX design prototyping	15	T2, T4	Ayham (UI/UX designer)	Figma

T6	API development	20	T3	Yazan, Ibrahim (software developers)	
T7	Frontend development	25	T5	Yazan, Ibrahim (software developers)	Html,Css and JavaScript
T8	System integration	15	T6, T7	Yazan, Ayham (integration specialist)	None
T9	Initial testing	20	T8	Ibrahim (Quality assurance tester)	
T10	User acceptance testing	10	T8	All team members	Zephyr
T11	Deployment planning	5	T9, T10	Yazan (Project manager)	None

T12	Deployment and monitoring	7	T11	All team members	None
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1.3.5 Legal Feasibility

A. Data Protection and Privacy

- Follow Jordanian data protection laws.
- Secure patient data with encryption.

B. Healthcare Regulations

- Meet Jordanian healthcare standards.
- Verify all listed doctors are licensed.

C. Intellectual Property

- Use properly licensed software.
- Secure trademarks for EasyAppointment.

D. Contracts and Agreements

- Set clear terms with healthcare providers.

- Create user agreements and privacy policies.

E. Liability and Risk

- Define medical liability and ensure insurance.
- Manage risks with regular policy reviews.

The Ministry of Health's adoption of EasyAppointment will support legal compliance and ensure the project meets all necessary regulations.

1.4 Recommended Solution and Expected Project Deliverables

Recommended Solution

The EasyAppointment project will provide a comprehensive healthcare and appointment scheduling system aimed at improving access to medical services across Jordan. The solution involves:

1. **Unified Platform:** A single system for searching doctors and booking appointments, integrating both public and private healthcare facilities.
2. **Medical ID System:** A secure digital ID for each user that stores medical history and personal information, ensuring doctors have all relevant details before consultations.
3. **User-Friendly Interface:** Intuitive web and mobile applications for easy navigation and appointment management.

4. **Integration with Healthcare Providers:** Connecting with both public and private healthcare providers to provide a complete directory of available doctors and services.
5. **Secure Data Management:** Implementing strong encryption and security measures to protect user data and ensure compliance with data protection laws.

Expected Project Deliverables

Functional Web Application:

- **Features:** Search for doctors, view doctor profiles, schedule and manage appointments, and access medical history.
- **Platform:** Accessible from desktop and mobile devices.

Mobile Application:

- **Features:** Similar to the web application, with additional mobile-specific functionalities.
- **Platforms:** Available for iOS and Android.

Medical ID System:

- **Features:** Secure digital profiles for users, including medical history and personal information.
- **Integration:** Linked to user accounts and accessible by healthcare providers during consultations.

Healthcare Provider Integration:

- **Features:** Seamless integration with public and private healthcare facilities, including real-time availability and booking.

Data Security Measures:

- **Features:** Encrypted data storage and secure data transmission.
- **Compliance:** Adheres to Jordanian data protection laws.

Training and Support Materials:

- **Documentation:** User guides, FAQs, and training materials for both users and healthcare providers.
- **Support:** Ongoing technical support and maintenance.

By delivering these components, the EasyAppointment project aims to enhance the accessibility and efficiency of healthcare services in Jordan, providing users with a reliable and user-friendly platform for managing their medical needs.

1.5 Local and Global Impact of the Proposed Solution

2.0 Improved Healthcare Access:

- **For Jordanian Citizens:** EasyAppointment will make it easier for people to find and access healthcare services, regardless of whether they are in public or private facilities. This improved access

can lead to better health outcomes and more efficient use of healthcare resources.

3.0 **Enhanced Efficiency:**

- **Appointment Management:** The system streamlines appointment scheduling, reducing waiting times and administrative burdens on both patients and healthcare providers.
- **Medical Information:** The Medical ID system ensures that doctors have immediate access to patient history, improving the quality of care and reducing errors.

4.0 **Support for Public Healthcare System:**

Integration with Public Providers: By including both public and private healthcare providers, the system helps balance the load across different

2.0 Project Management plan

2.1 Project Organization

The software documentation will go as follows (including the ones above):

- ❖ **Chapter 3:** Software Requirements Specifications (SRS)
Stakeholder roles, technical and performance requirements, software needs, and system architecture.
- ❖ **Chapter 4:** Analysis and Design
Activity Diagrams, Sequence Diagrams, Class Diagram, Architecture Design, Classes and Components Design, Graphical User Interface Design, Object to ER Mapping and Physical Database Design (if required)
- ❖ **Chapter 5:** Implementation
Programming languages used, documentation of testing procedures and test cases.
- ❖ **Chapter 6:** User Manual
We will Provides instructions for users to know how to use our software.
- ❖ **Chapter 7:** References: books and tools

2.2 Roles and Responsibilities

Table 4: roles and responsibilities

Name	Role	Responsibilities
Yazan Jaber	Project Manager & project Architect	Plan, execute, and finalize the project on time and within budget. Distribute tasks appropriately. Manage resources and ensure quality. Define the technical and functional architecture of the system. Develop the most critical components of the system. Perform feasibility studies. ensuring adherence to architectural standards.
Ayham Obeidat	UI/UX designer	Responsible for the design of the prototypes and the GUI. Create visually appealing and user-friendly designs. Ensure consistency and usability throughout the project.
Ibrahim Abu Jaber	Developer	Responsible for developing and writing code for assigned tasks and testing it thoroughly to make sure it meets requirements and quality standards.

2.3 Software Process Mode

The goal is to develop a comprehensive healthcare system website with diverse, independent features. Community feedback is important; however, for this project, we will follow a Waterfall development approach. This means that each phase of the project will be completed sequentially, with clearly defined milestones and deliverables at each stage. Changes to requirements will be managed through formal change control processes, and the project

will proceed through a structured series of phases including requirements gathering, design, development, testing, and deployment.

2.4 Tools and Techniques

Tools:

In the development of our website, we will leverage key programming languages including HTML, CSS, and JavaScript within the widely recognized web development framework, Express.js. Database management will be assigned to MySQL. Our design process will be facilitated by Figma, serving dual roles in prototyping, and refining the UI/UX design.

Techniques:

In our team, individuals bring valuable expertise from previous projects, and we will leverage this experience to gauge the effort required for the development of this project. Additionally, we plan to employ the state transition technique for testing our website.

Lastly, we will carry out several reviews along the way to ensure and evaluate the ongoing quality of our work.

2.6 Assigning Team Members to Tasks

Project manager: Yazan Jaber.

Software developers: Yazan Jaber, Ibrahim Abu Jaber.

UI/UX Designer: Ayham OBEIDAT.

Quality Assurance (QA) Tester: Ibrahim Abu Jaber.

Business analyst: Ayham OBEIDAT.

System architect: Yazan Jaber.

System Design Team: Yazan Jaber, Ibrahim Abu Jaber.

Integration specialist: Yazan Jaber, Ayham OBEIDAT.

2.5 and 2.7 - Work breakdown and project Schedule (Gantt chart and PERT diagram)

Table 5: work breakdown and project Schedule

Task ID	Task Description	Duration	Dependencies	Responsible staff	resources
T1	Initial Stake Holder meeting	5	None	All team member	None
T2	Requirements specifications	7	T1 (M1)	Ayham (BA)	None
T3	System Architecture design	14	T2 (M2)	Yazan (System Architect)	Lucid chart
T4	Detailed system design	12	T3 (M3)	Yazan, Ibrahim (system design team)	starUML
T5	UI/UX design prototyping	15	T2, T4 (M4)	Ayham (UI/UX designer)	Figma

T6	API development	20	T3 (M4)	Yazan, Ibrahim (software developers)	
T7	Frontend development	25	T5 (m5)	Yazan, Ibrahim (software developers)	Swift, Kotlin
T8	System integration	15	T6, T7 (M5)	Yazan, Ayham (integration specialist)	None
T9	Initial testing	20	T8 (M7)	Ibrahim (Quality assurance tester)	
T10	User acceptance testing	10	T8 (M7)	All team members	Zephyr
T11	Deployment planning	5	T9, T10 (M8)	Yazan (Project manager)	None

T12	Deployment and monitoring	7	T11 (M9)	All team members	None
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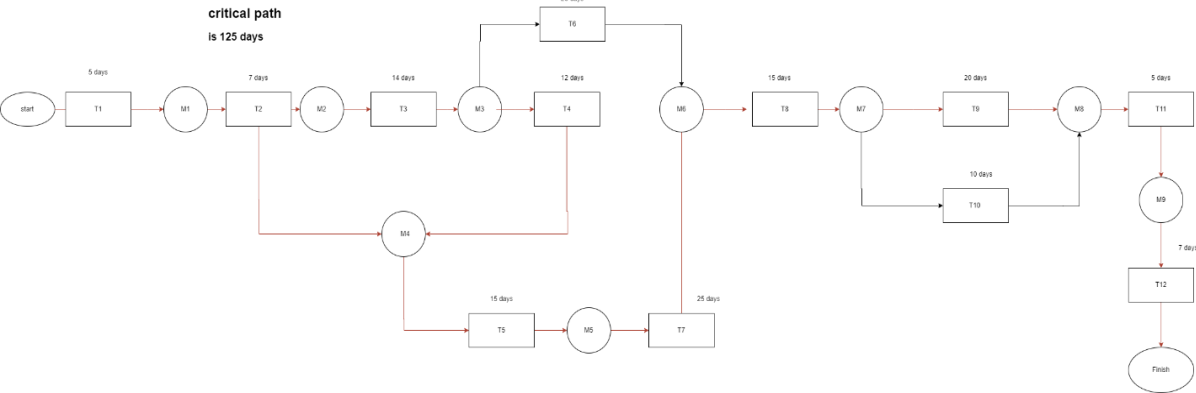


Figure 1: pert diagram

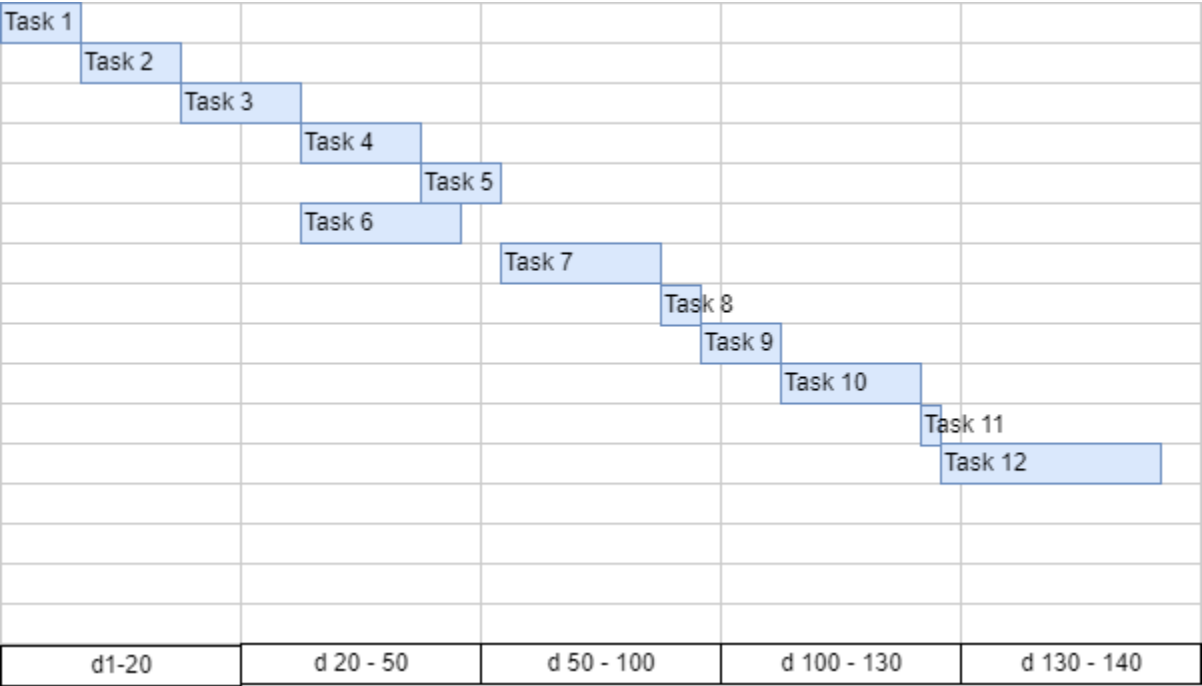


Figure 2: Gantt chart

2.8 Risk Analysis and Plans

1 – Infrastructure challenges: the app/website would need to accommodate both digital and paper-based medical records, which can be challenging when ensuring all relevant patient information (the patient's name, contact, age, and medical history which includes any diseases, medications and allergies) is accessible and up to date.

2 – Digital literacy: Not all patients or healthcare professionals may be familiar with using such mobile apps, requiring education and training to ensure effective utilization of the system.

3 – Cultural sensitivity: the app/website content and features may not be socially acceptable in respect with local customs and beliefs, such as women/pregnancy related information or mentally issues information that some people in the public may find sensitive information.

4 – Financial support: finding stakeholders who will fund the development, maintenance, and implementation the whole way through until the final product may be challenging.

5 – Marketing/Promotion: the app would need an effective marketing campaign to allow for the app to be widespread amongst the Jordanian public to fully utilize the apps capabilities.

2.9 Monitoring, Reporting, and Controlling Mechanisms

We will be monitoring the app during the development (through daily meeting in the morning, weekly reports of what everyone did and what problems they faced, performance metrics such as tracking KPIs which are task completion rates, defect counts, and resource utilization which they can be presented every month) of it

And after the release date to the public (through having in the app an issues box that the user can write what issues they faced with

the app and having a contact number or an email so the user can
write what problem they are facing)

3.0 Software Requirements

Specifications (SRS)

3.1 System Stakeholders and Requirements Sources (Identifies stakeholders and sources of requirements)

Table 6: stakeholders

Stake holder	Description
Patients	Individuals who will use the app to access health care services
Healthcare provider	Doctors, nurses, and other medical staff who will use the app
Jordan ministry of health	As the client or a main sponsor
IT development team	The group of people who will develop, maintain, test the app, and building it
Healthcare facility administrator	Individuals responsible for managing the profiles of the patients who will be using the app within the facility

We will be able to identify our requirements by multiple different ways

- 1- **Surveys:** it will have questions meant to be answered by users and stakeholders about what they want in the app and what is there preferences and needs for the app.

- 2- **Interviews:** will be held one-on-one discussions with stakeholders, patients, administrators, healthcare providers, about what they want with the app what they like what they don't like and what they would like to add.
- 3- **Ethnography:** make one of the working team go to an actual hospital/clinic to observe how workers behave with each other and communicate to know what the requirements are needed for this project.
- 4- **User stories and use cases:** narrative description that focuses on the user's perspective and describes the tasks they want to perform with the app, they help ensure that the app development align with real user's needs.

3.2 User Requirement Definition

The app should have all the basic needs for the user to use it with its functional, non-functional, interface requirements.

Functional:

- the app must include a login/signup page for all user
- the app must include an updating profile mechanism (updating the contact, password, etc.)
- the app must include a stay signed in mechanism to make it easy for the user to get into the app.
- The app should have a different page for logging in for healthcare providers.
- The app should enable the user to input his medical history.

Non-Functional:

- The app should ensure that the user's data is safe by implementing SSL, data encryption, and complying with data privacy laws.
- The app's interface should be easy to use for all people despite their age or education.
- The app should be quick.

- The app should be available 24/7.
- code should be well documented for easy change.
- The app should work across different platforms and devices.

3.3 Use case Diagrams

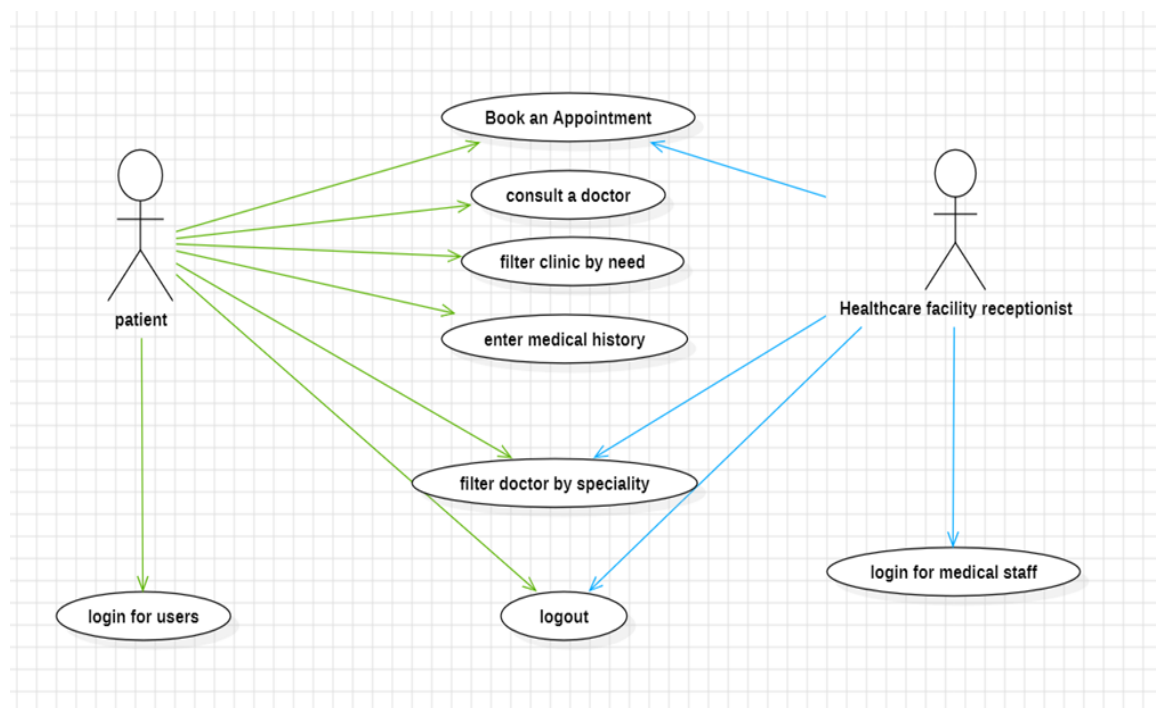


Figure 3: use case diagram for patient and healthcare facility receptionist

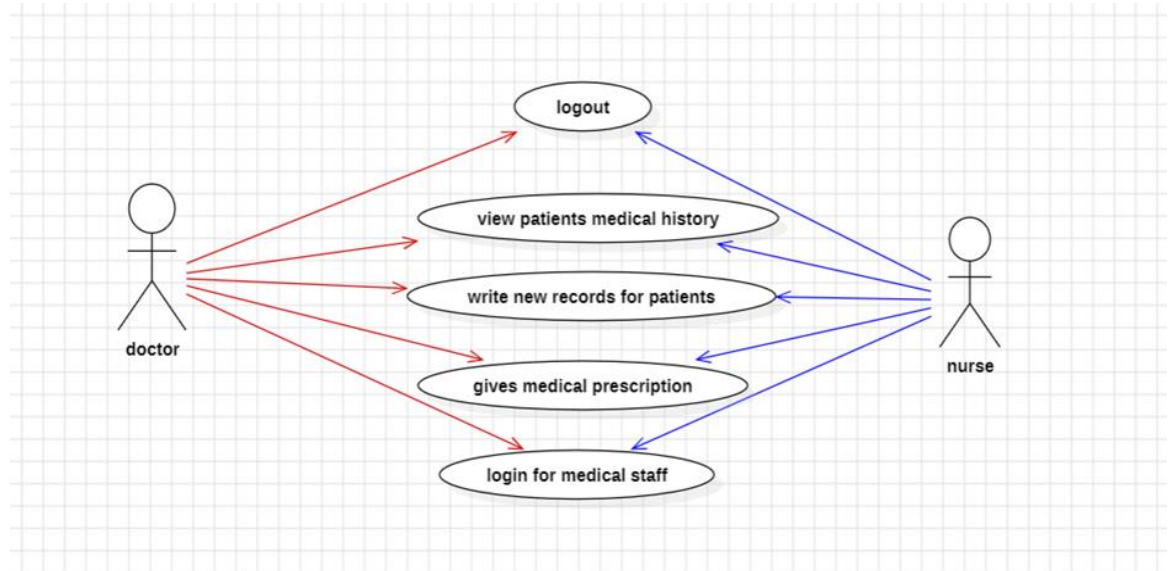


Figure 4:use case diagram for doctor and nurse

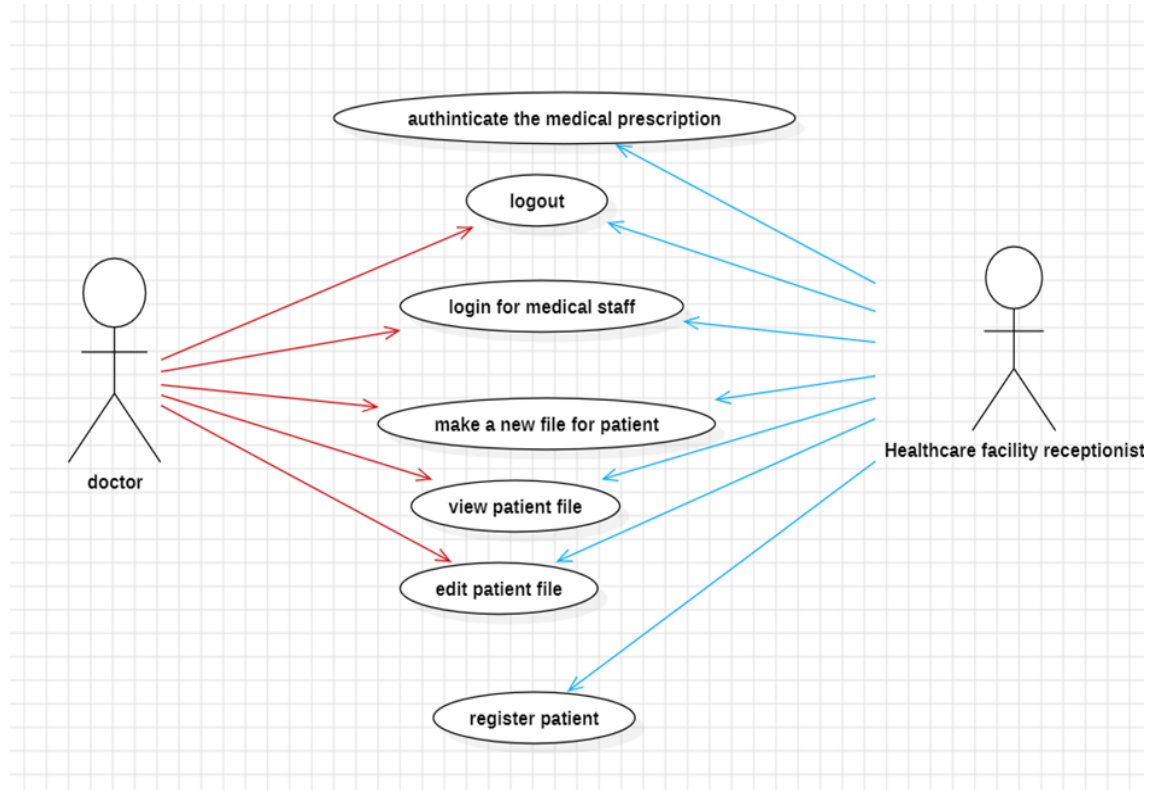


Figure 5: use case diagram for doctor and healthcare facility receptionist

3.4 System Functional Requirement Specification

User Stories

1. Patient User Registration: As a new patient, I want to register for an account on the EasyAppointment platform so that I can access healthcare services and manage my appointments.
2. Healthcare Provider Registration: As a healthcare provider, I want to register my practice on the EasyAppointment platform so that patients can find me and book appointments.
3. Personal Health Record (PHR) System: As a patient, I want to access and manage my Personal Health Record (PHR) so that I can track my medical history and share it with healthcare providers.

4. Search for a Doctor: As a patient, I want to search for doctors based on my location, specialty, and financial capacity so that I can find the best available healthcare provider for my needs.
5. Book an Appointment: As a patient, I want to book an appointment with a doctor so that I can schedule a consultation at a convenient time.
6. View Medical History: As a doctor, I want to access a patient's medical history so that I can provide informed medical care.
7. Manage Appointments: As a healthcare provider, I want to manage and update my appointment schedule so that I can optimize my availability and patient flow.
8. Patient Profile Management: As a patient, I want to update my personal and medical information so that my records are accurate and up to date.

Requirement Scenarios

1. Patient Registration: Given a new user is accessing the system, when they choose to register as a patient, then the system should guide them through the registration process, collecting necessary information and creating an account.
2. Healthcare Provider Registration: Given a healthcare provider is accessing the system, when they choose to register their practice, then the system should collect details about the provider and their services and create a profile for them on the platform.
3. Access PHR: Given a patient wants to view their health records, when they navigate to the PHR section, then the system should display their medical history, treatments, and other health-related data.
4. Search for a Doctor: Given the patient is logged into the system, when they enter search criteria (location, specialty, financial range), then the system should display a list of doctors that match the criteria.

5. Book an Appointment: Given a patient has selected a doctor, when they choose an available time slot, then the system should confirm the booking and send a notification to both the patient and the doctor.
6. View Medical History: Given a doctor is viewing a patient's profile, when they click on the medical history tab, then the system should display the patient's past consultations, treatments, and any uploaded medical documents.
7. Manage Appointments: Given a healthcare provider wants to view their schedule, when they log in to the system, then they should be able to see all upcoming appointments and make changes as necessary.
8. Update Patient Profile: Given a patient needs to update their information, when they navigate to the profile section, then the system should allow them to edit and save their personal and medical details.

3.5 Textual Description for Each Use Case

Case

This section provides a detailed textual description for each use case identified in the EasyAppointment system.

Use Case 1: Patient User Registration

- Primary Actor: Patient
- Goal: Create an account on the EasyAppointment platform.
- Preconditions: The user has access to the internet and the EasyAppointment platform.
- Main Success Scenario:
 - The patient accesses the registration page and fills in personal details, including name, contact information, and medical history.

- The system creates a new patient account and sends a confirmation email.
- Postcondition: The patient has a registered account and can access the system's features.

Use Case 2: Healthcare Provider Registration

- Primary Actor: Healthcare Provider
- Goal: Register their practice on the EasyAppointment platform.
- Preconditions: The provider has access to the internet and the EasyAppointment platform.
- Main Success Scenario:
 - The provider accesses the registration page for healthcare providers and fills in details about their practice, including services offered, location, and contact information.
 - The system creates a provider profile and makes it searchable by patients.

- Postcondition: The provider's profile is active on the platform, allowing patients to find and book appointments.

Use Case 3: Personal Health Record (PHR) System

- Primary Actor: Patient
- Goal: Access and manage their Personal Health Record (PHR).
- Preconditions: The patient is logged into the system and has an existing PHR.
- Main Success Scenario:
 - The patient navigates to the PHR section, and the system displays the patient's medical history, including previous consultations, treatments, and uploaded documents.
 - The patient can update or add information to their PHR.
- Postcondition: The patient's PHR is up-to-date and accessible.

Use Case 4: Search for a Doctor

- Primary Actor: Patient
- Goal: Find a doctor based on specific criteria such as location, specialty, and financial capacity.
- Preconditions: The patient is logged into the system.
- Main Success Scenario:
 - The patient navigates to the search page and enters search criteria (location, specialty, financial range).
 - The system displays a list of matching doctors.
 - The patient selects a doctor from the list for further action (e.g., booking an appointment).
- Postcondition: The patient finds a suitable doctor for their needs.

Use Case 5: Book an Appointment

- Primary Actor: Patient

- Goal: Schedule an appointment with a doctor.
- Preconditions: The patient has selected a doctor.
- Main Success Scenario:
 - The patient views the doctor's available time slots, selects a convenient time, and confirms the appointment.
 - The system sends a confirmation notification to both the patient and the doctor.
- Postcondition: The appointment is successfully booked.

Use Case 6: View Medical History

- Primary Actor: Doctor
- Goal: Access and review a patient's medical history.
- Preconditions: The doctor is logged into the system and has the necessary permissions.
- Main Success Scenario:

- The doctor navigates to the patient's profile and selects the medical history tab.
- The system displays the patient's past consultations, treatments, and medical documents.
- Postcondition: The doctor has a comprehensive view of the patient's medical history.

Use Case 7: Manage Appointments

- Primary Actor: Healthcare Provider
- Goal: Manage and update their appointment schedule.
- Preconditions: The healthcare provider is logged into the system.
- Main Success Scenario:
 - The provider navigates to the appointments page and views their upcoming appointments.
 - The provider can add, modify, or cancel appointments as needed.

- Postcondition: The provider's schedule is up-to-date and optimized.

Use Case 8: Patient Profile Management

- Primary Actor: Patient
- Goal: Update personal and medical information.
- Preconditions: The patient is logged into the system.
- Main Success Scenario:
 - The patient navigates to the profile section and edits their personal and medical details.
 - The system saves the changes.
- Postcondition: The patient's profile is accurate and up to date.

3.6 Non-Functional Requirements

In this section, I will outline the Non-Functional Requirements (NFRs) for the EasyAppointment system. These requirements will cover the essential aspects of performance, dependability, security, usability, operational conditions, and maintainability that the system must meet. Additionally, I will explain how each of these requirements can be tested after the project is completed, ensuring that the system functions as intended and meets the quality standards necessary for deployment in the healthcare sector.

3.6.1 Performance Requirements

The system must perform efficiently to ensure timely access to healthcare services for all users.

- **Response Time:** The system must retrieve and display search results (e.g., finding a doctor or medical facility) within 2 seconds, even during peak usage times.
- **Throughput:** The system should handle up to 1,000 concurrent users with no degradation in performance, ensuring smooth operation during high traffic.
- **Latency:** Data retrieval and submission (e.g., booking an appointment) should occur with a maximum latency of 1 second.
- **Scalability:** The system should be able to scale to accommodate a 20% increase in users each year without requiring major architectural changes.

3.6.2 Dependability Requirements

Dependability is critical to ensure that the system is reliable and available to users, given the importance of healthcare services.

- **Availability:** The system must have an uptime of 99.95%, ensuring it's available 24/7, especially during emergencies.
- **Reliability:** The system should operate without failure for an average of 6 months (MTBF - Mean Time Between Failures).
- **Fault Tolerance:** The system should automatically failover to a backup server in case of hardware or software failure, with no more than 30 seconds of downtime.
- **Recovery:** The system should recover from a failure within 10 minutes, ensuring minimal disruption to users.

3.6.3 Security Requirements

Security is paramount to protect sensitive medical information and ensure compliance with healthcare regulations.

- **Data Encryption:** All patient data must be encrypted using AES-256 encryption, both at rest and in transit, to protect against unauthorized access.
- **Access Control:** Implement multi-factor authentication (MFA) for all users accessing medical records, with role-based access controls to limit data exposure.
- **Audit Trails:** The system must log all access to patient data, including who accessed the data, what was accessed, and when, with logs retained for at least 5 years.

- **Vulnerability Management:** Conduct regular security audits and patch vulnerabilities within 48 hours of discovery to ensure the system remains secure.

3.6.4 Usability Requirements

The system should be easy to use, ensuring that all citizens, regardless of their technical skills, can navigate the system effectively.

- **User Interface Design:** The interface should be intuitive, with a minimal learning curve, enabling users to book appointments within 3 clicks.
- **Accessibility:** The system must be compliant with WCAG 2.1 AA standards, ensuring it is accessible to users with disabilities.
- **Training Requirements:** Provide online tutorials and help documentation in both Arabic and English, with a support hotline available for additional assistance.
- **Error Handling:** The system should display clear, user-friendly error messages, guiding users on how to correct mistakes or proceed with alternative actions.

3.6.5 Operational and Environmental Requirements

This section covers the operational context and environmental conditions under which the system will function.

- **Hardware Requirements:** The system should run on servers with a minimum specification of 16 GB RAM, 8-core CPU, and 500 GB SSD storage.
- **Software Dependencies:** The system must be compatible with major browsers (Chrome, Firefox, Edge) and run on Windows Server 2019 or later.
- **Environmental Conditions:** If deployed on premises, the system should operate within temperature ranges of 10°C to 35°C and relative humidity of 20% to 80%.
- **Backup and Disaster Recovery:** Daily backups should be performed, with data stored offsite. In case of disaster, the system should be fully restored within 4 hours.

3.6.6 Maintainability Requirements

Maintainability is essential for the long-term success of the system, allowing for easy updates and modifications.

- **Modularity:** The system should be designed with modular components, allowing individual modules (e.g., appointment booking, patient records) to be updated independently.
- **Documentation:** Comprehensive documentation must be provided, including API documentation, user manuals, and maintenance guides, updated with every major release.
- **Update Process:** System updates should be deployable with zero downtime, following a continuous integration/continuous deployment (CI/CD) pipeline.
- **Supportability:** The system should include built-in monitoring tools to detect and report issues, with a support team available 24/7 to address critical problems.

3.7 data requirements

Data entities and attributes

Patient information:

Entity: patient

Attributes: name, age, medical history, email, phone number.

Medical staff information:

Entity: medical staff

Attributes: name, age, id, email, phone number, specialty

clinics/ hospitals information:

entity: clinics/ hospitals

attributes: contact information, location, working times, number of staff, number of rooms, max capacity for patients

data validation rules

email validation:

rule: the email should take the normal format which is ([A@B.C](#))

medical staff validation:

rule: the staff can't login without their medical ID or their card

patient validation:

rule: the patient must enter his medical history and what type of doctor he needs before making an appointment

data storage and retrieval

database structure:

requirement: store the data entities using a relational database structure with appropriate tables, columns, primary/foreign key relationships.

Indexing:

Requirement: for every new user he is given a unique id.

Data access methods:

Requirement: we should enable the create, edit, delete, and update methods for efficient data retrieval and manipulation

Query performance:

Requirement: ensure database queries are optimized and run efficiently

4.0 Analysis and Design

4.1 Activity Diagrams

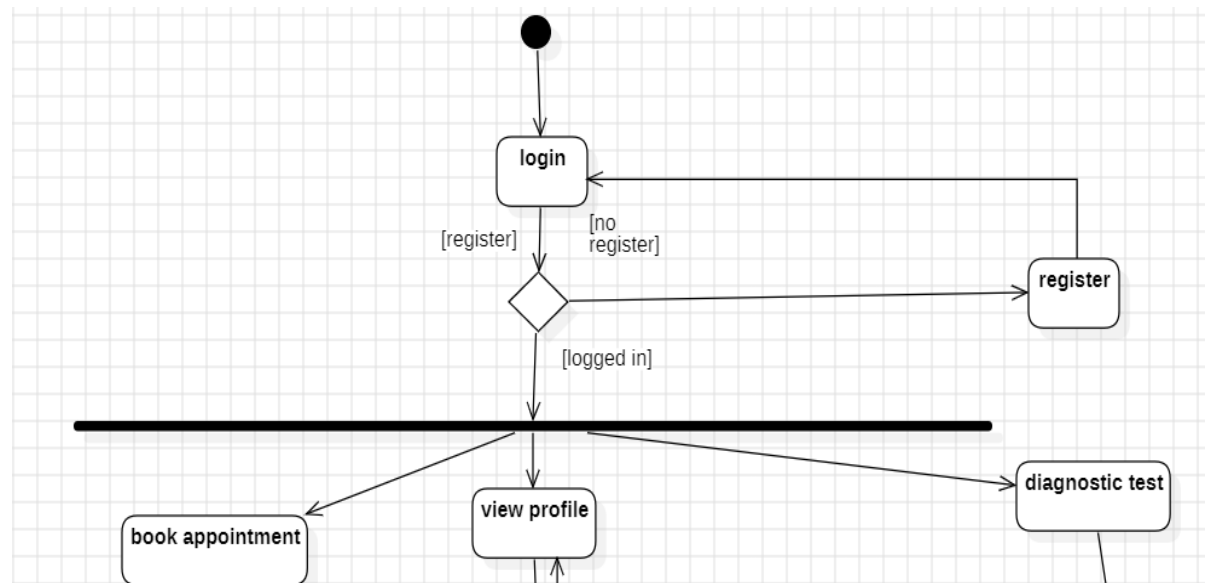


Figure 6: head of the activity diagram for a patient

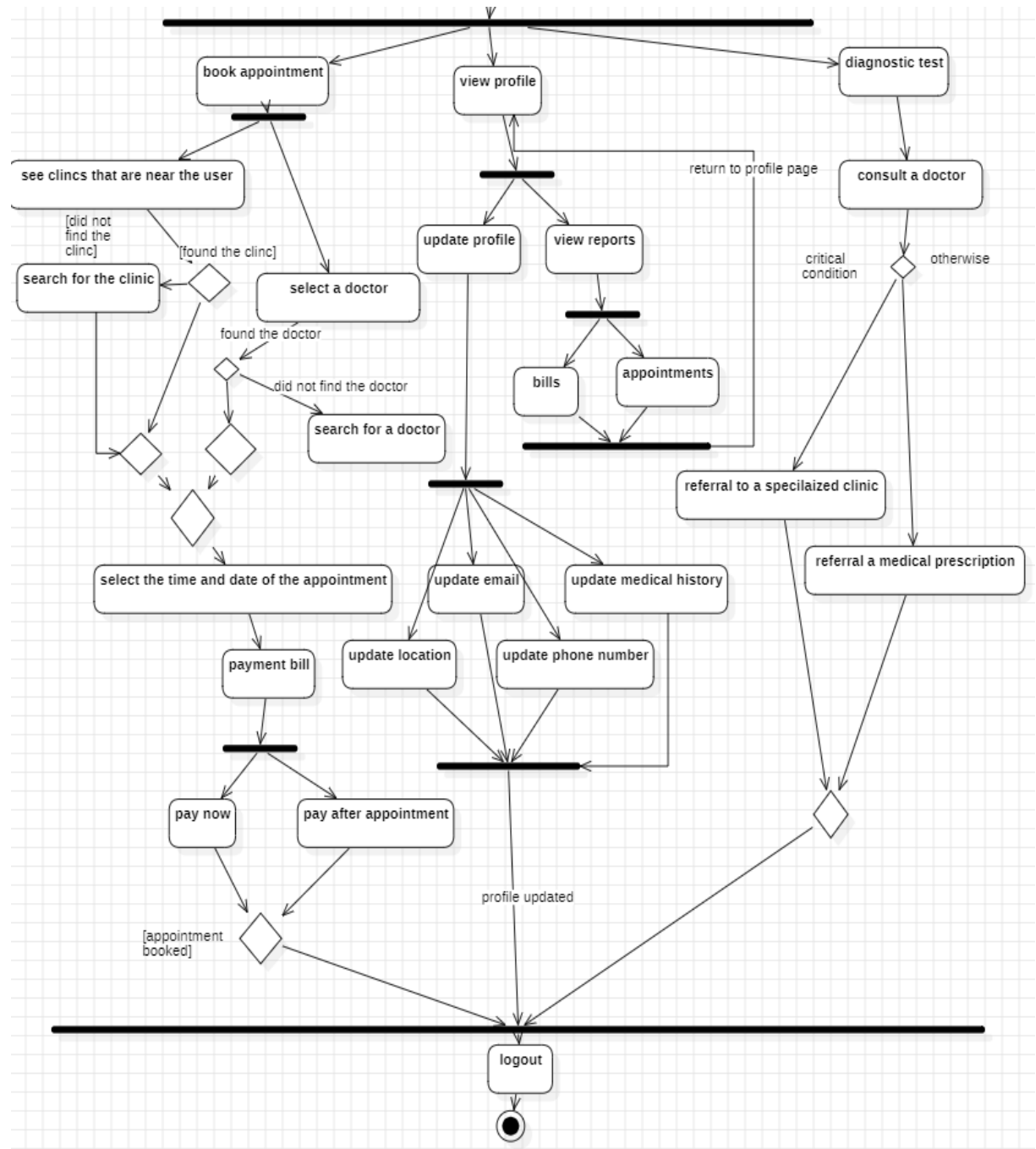


Figure 7: the body of the activity diagram for a patient

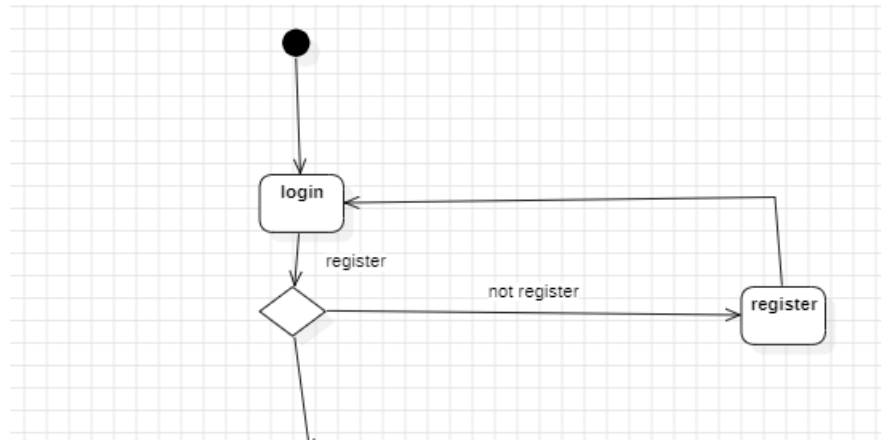


Figure 8: head of the activity diagram for a doctor

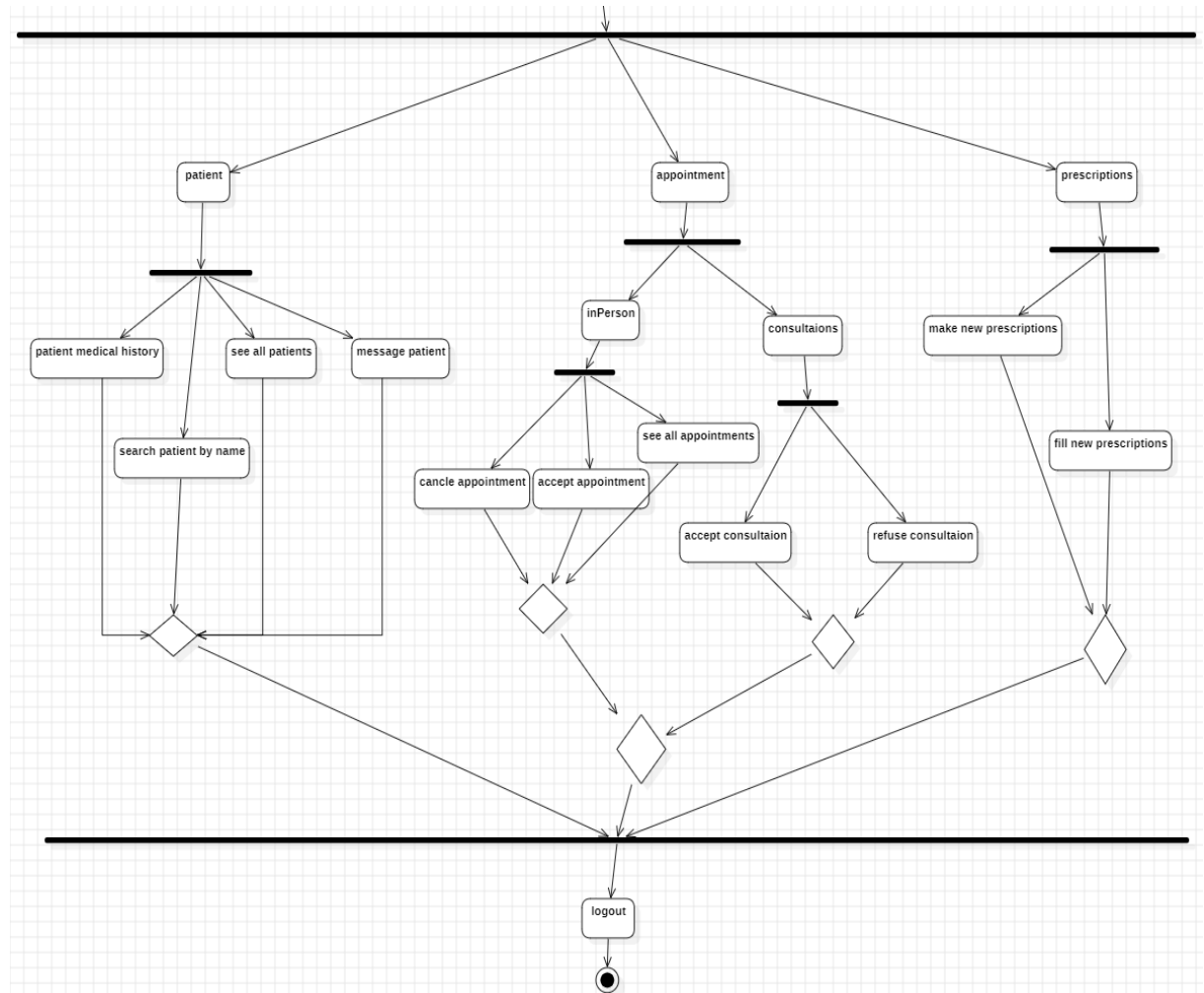


Figure 9: body of the activity diagram for a doctor

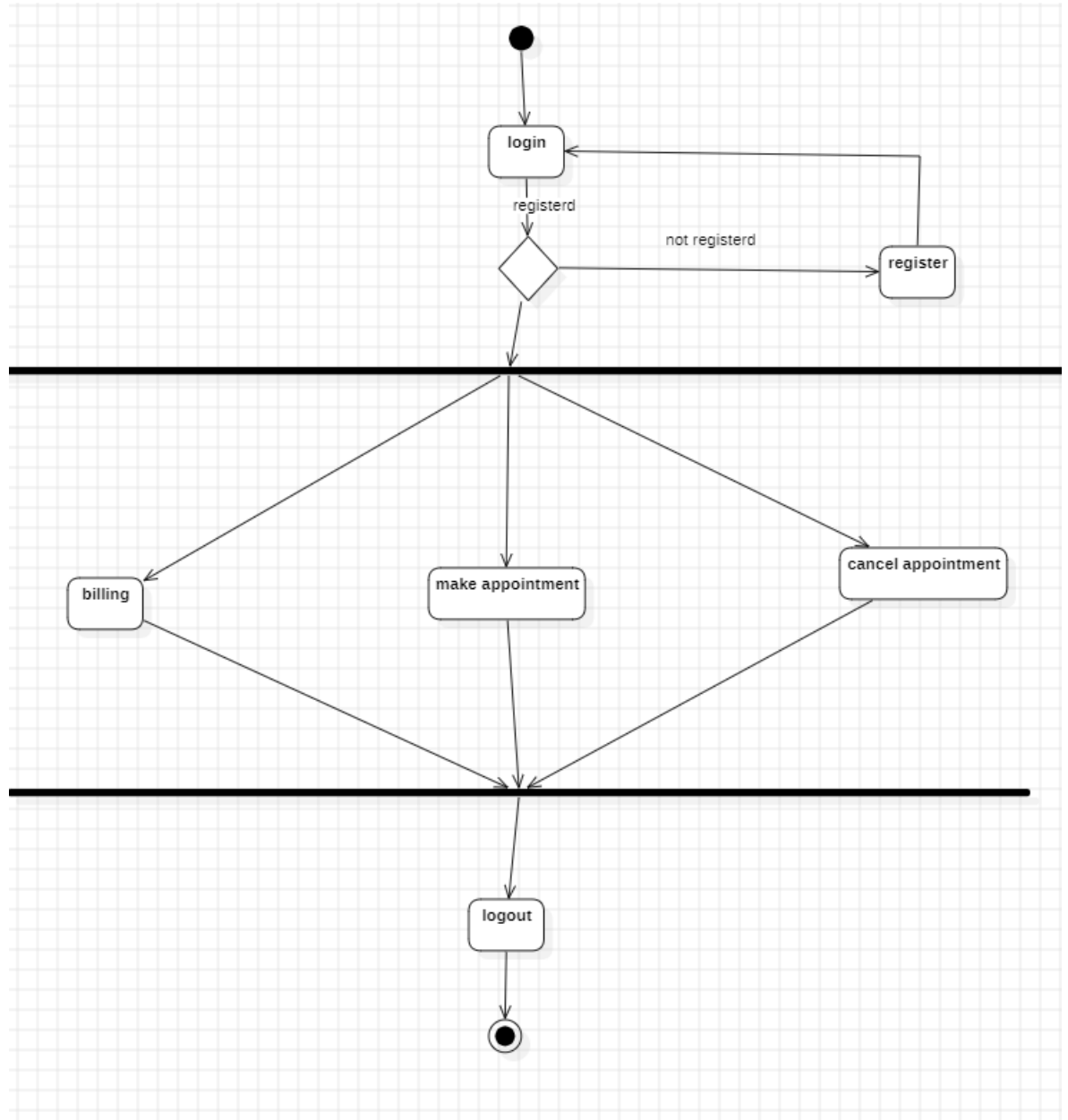


Figure 10: activity diagram for a healthcare facility receptionist

4.2 Sequence Diagrams

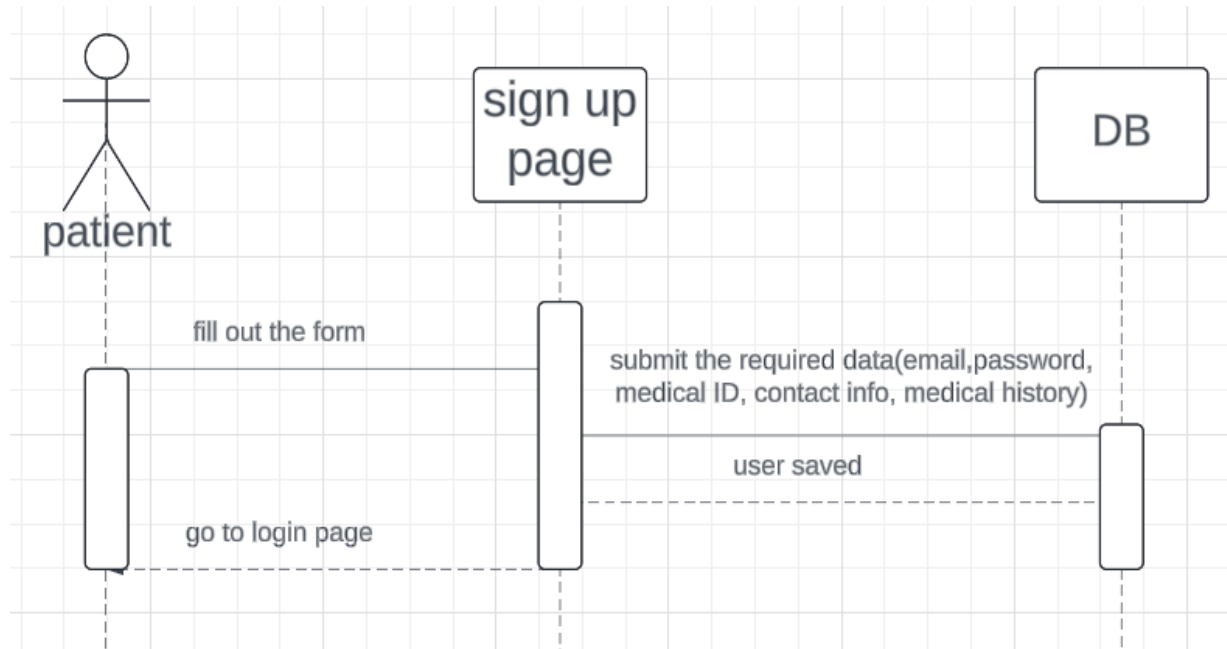


Figure 11: sequence diagram for patient registration

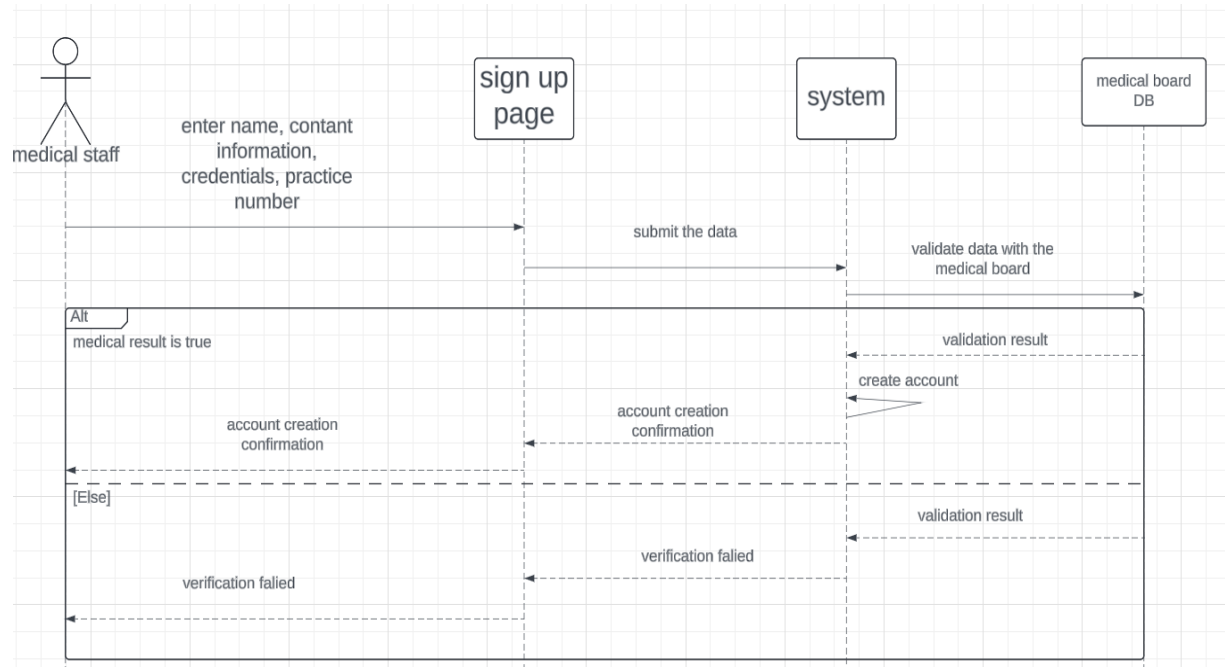


Figure 12: sequence diagram for medical staff registration

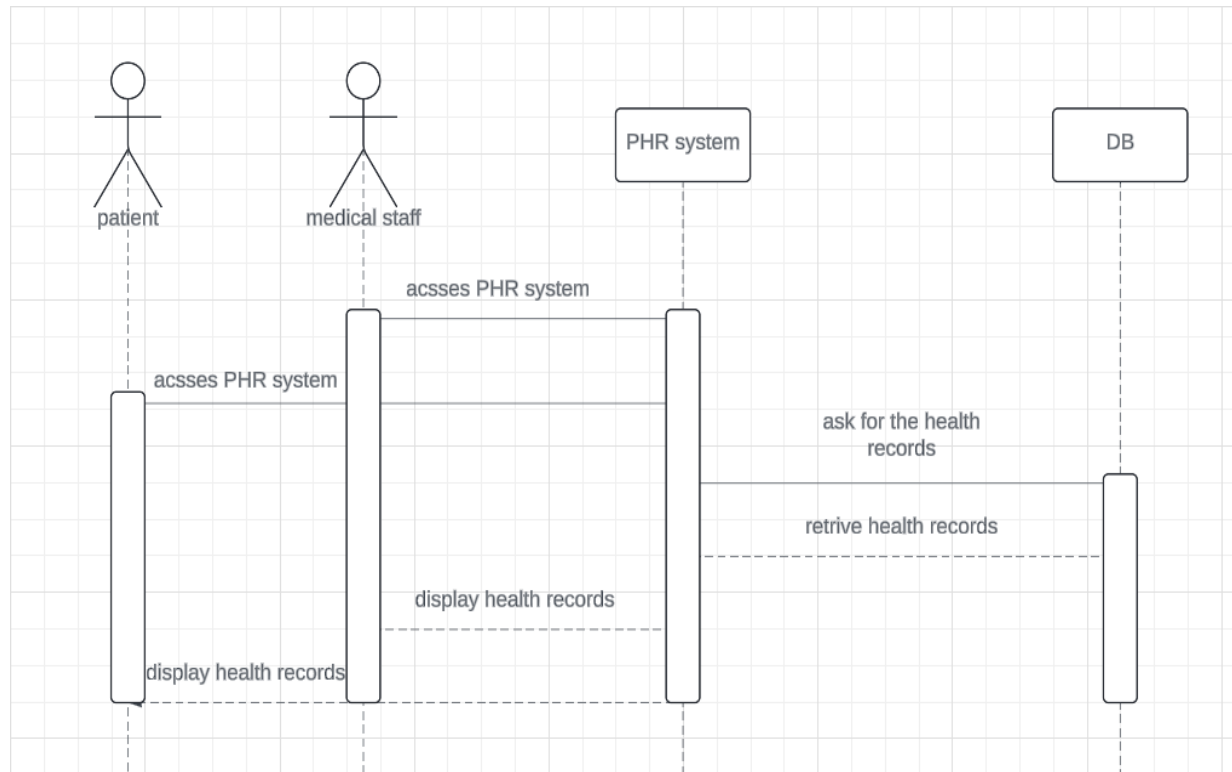


Figure 13: sequence diagram for PHR system

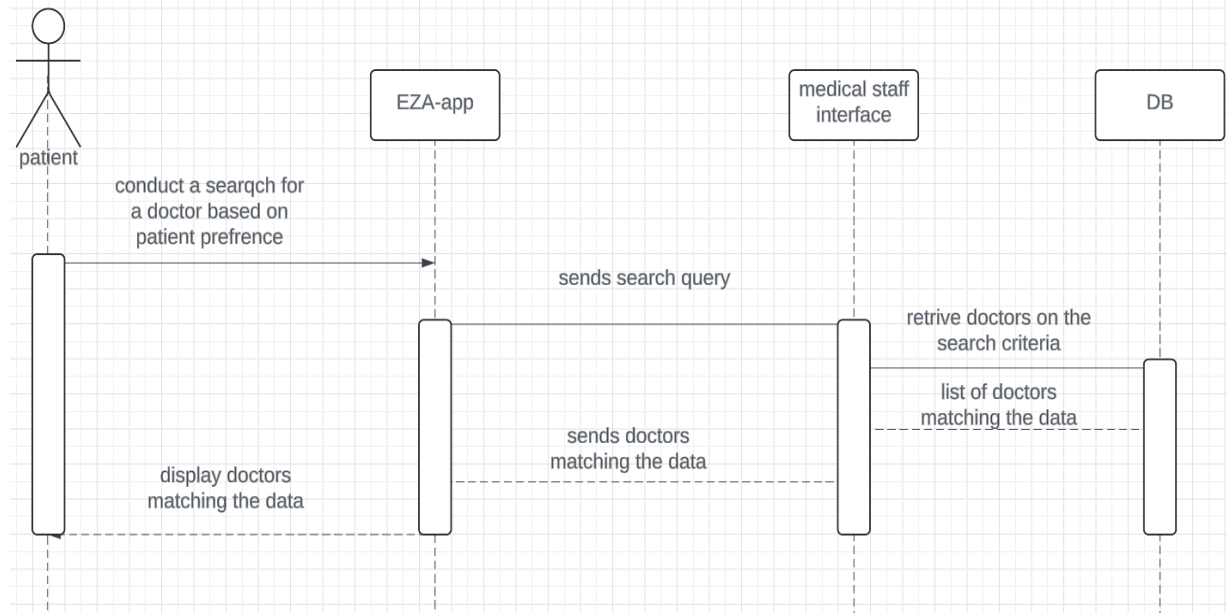


Figure 14: sequence diagram for searching for a doctor

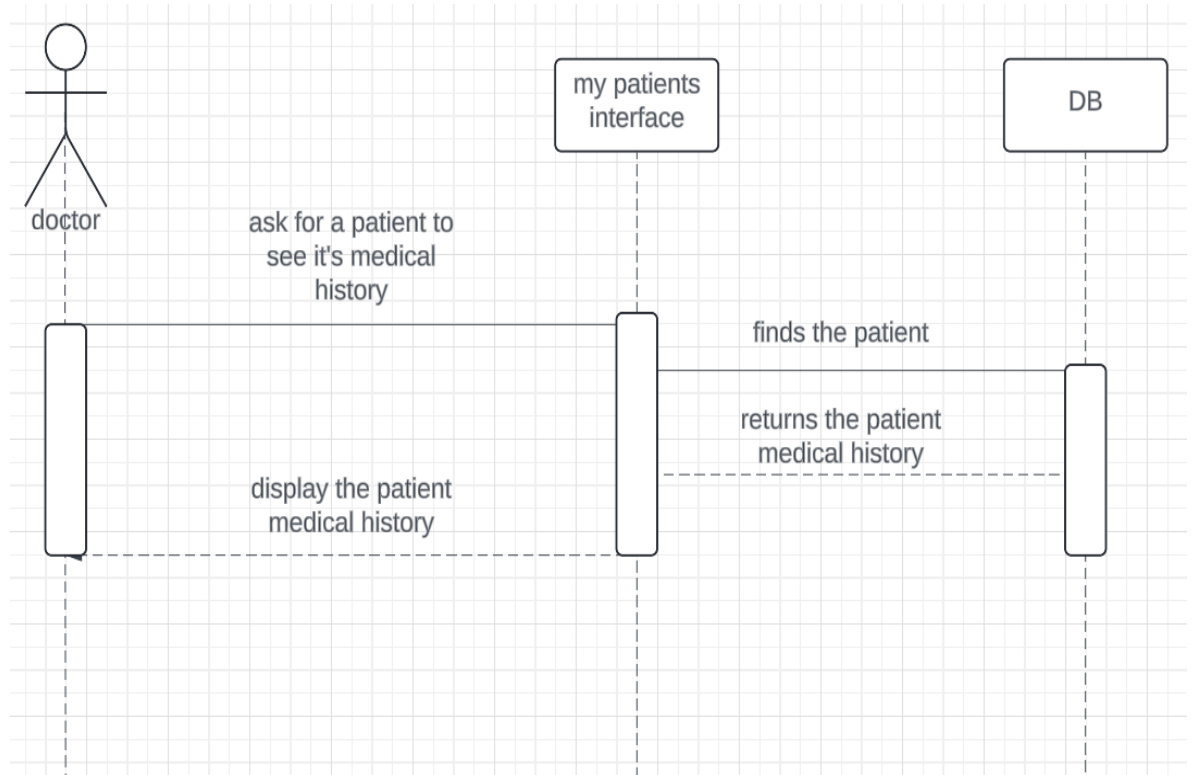


Figure 15: sequence diagram for seeing a patient medical history

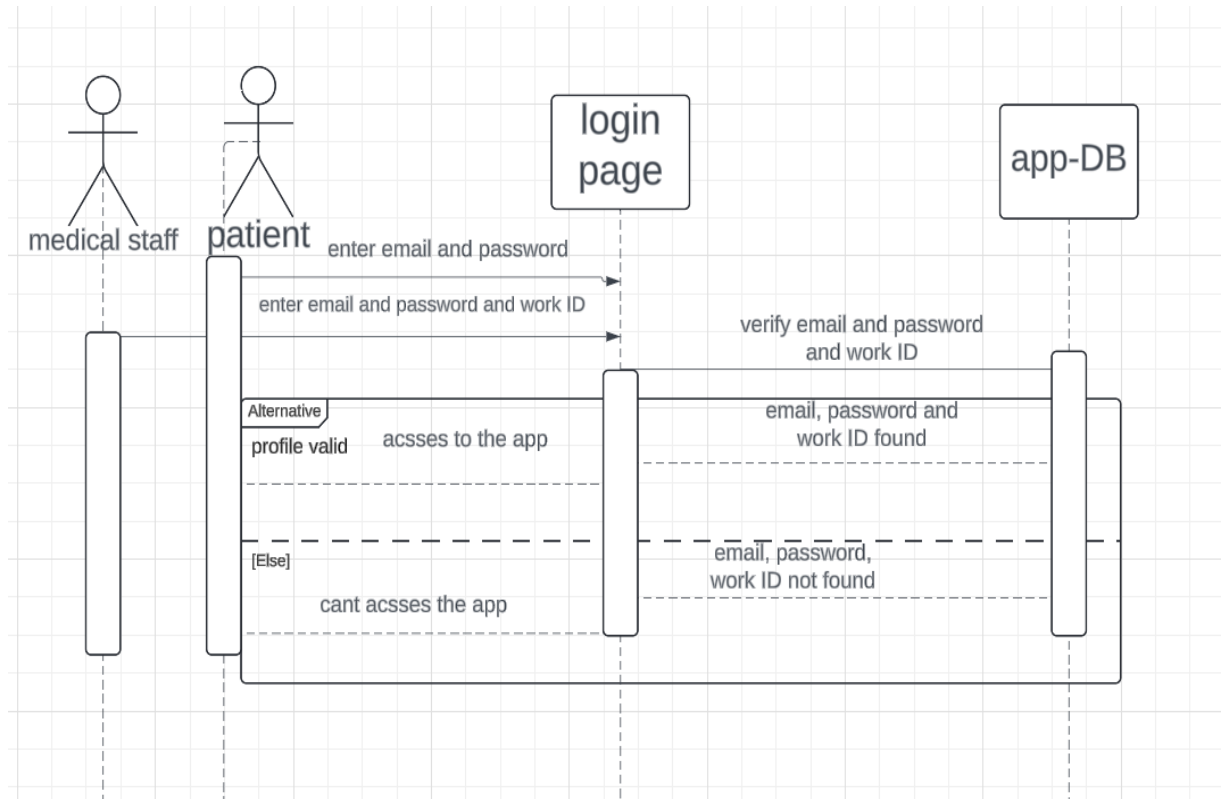


Figure 16: sequence diagram for logging in

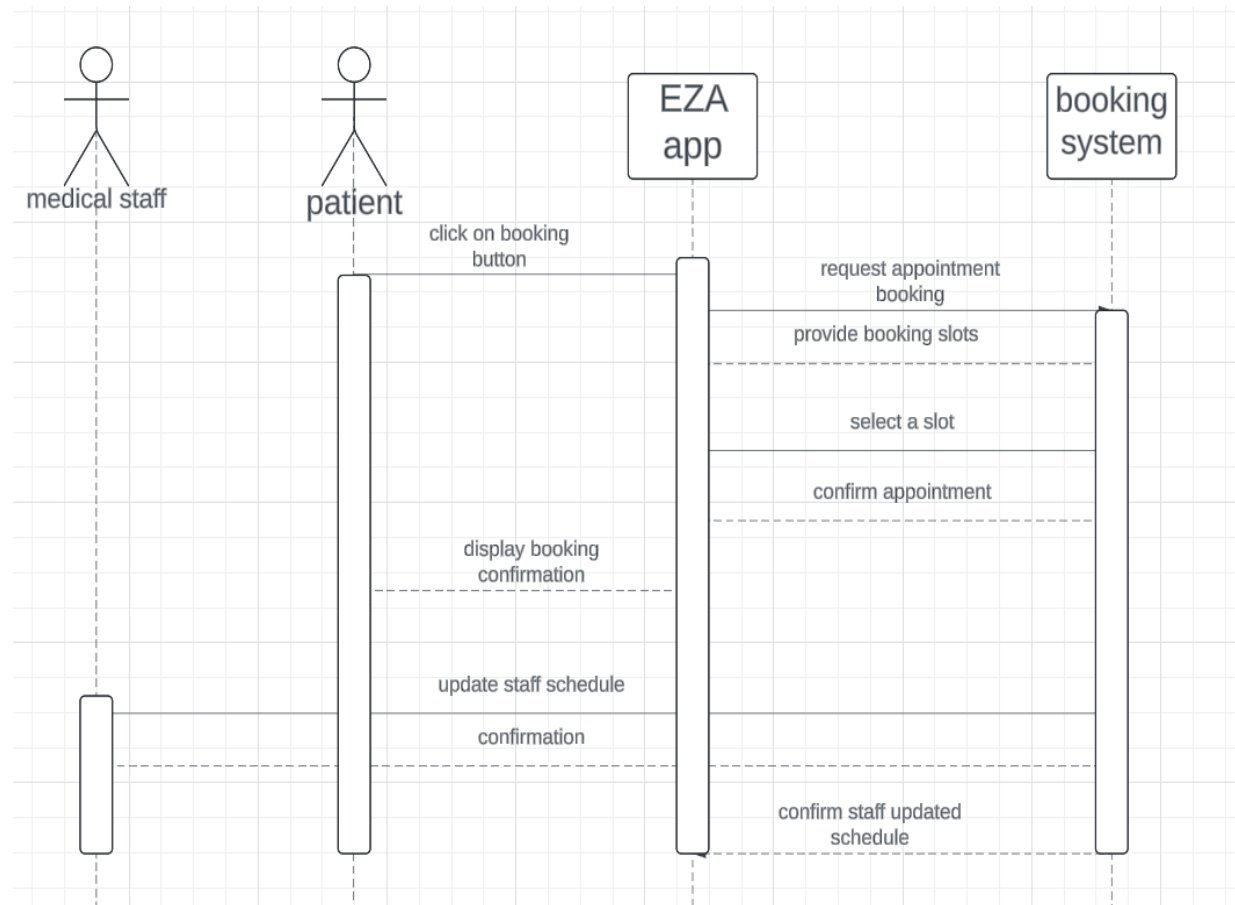


Figure 17: sequence diagram for booking an appointment

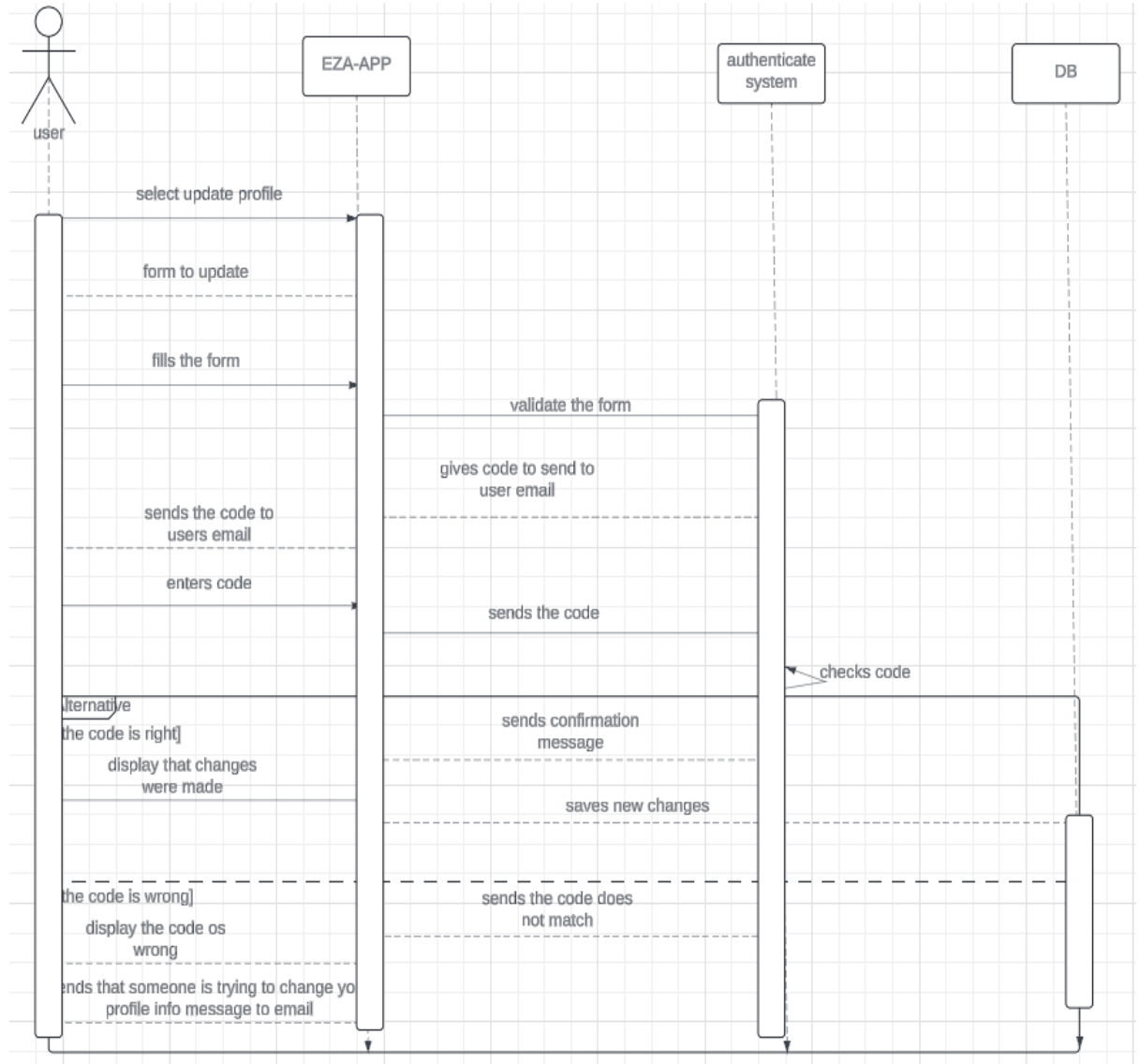


Figure 18: sequence diagram for updating user profile

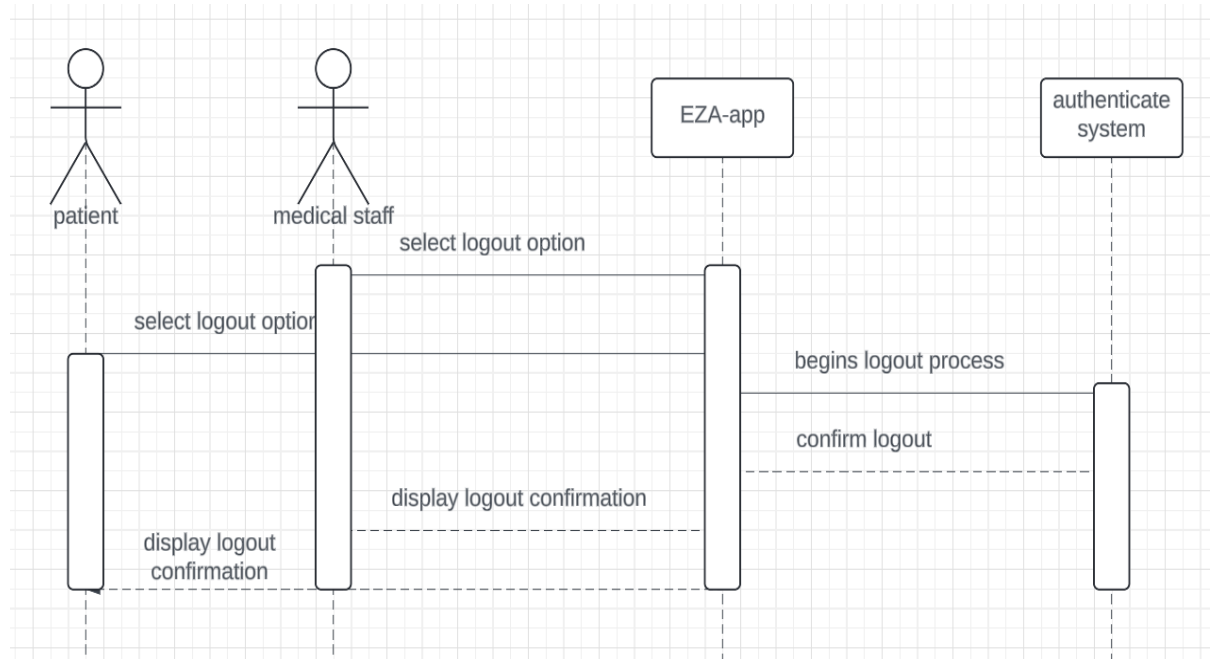


Figure 19: sequence diagram for logging out

4.3 Class Diagram

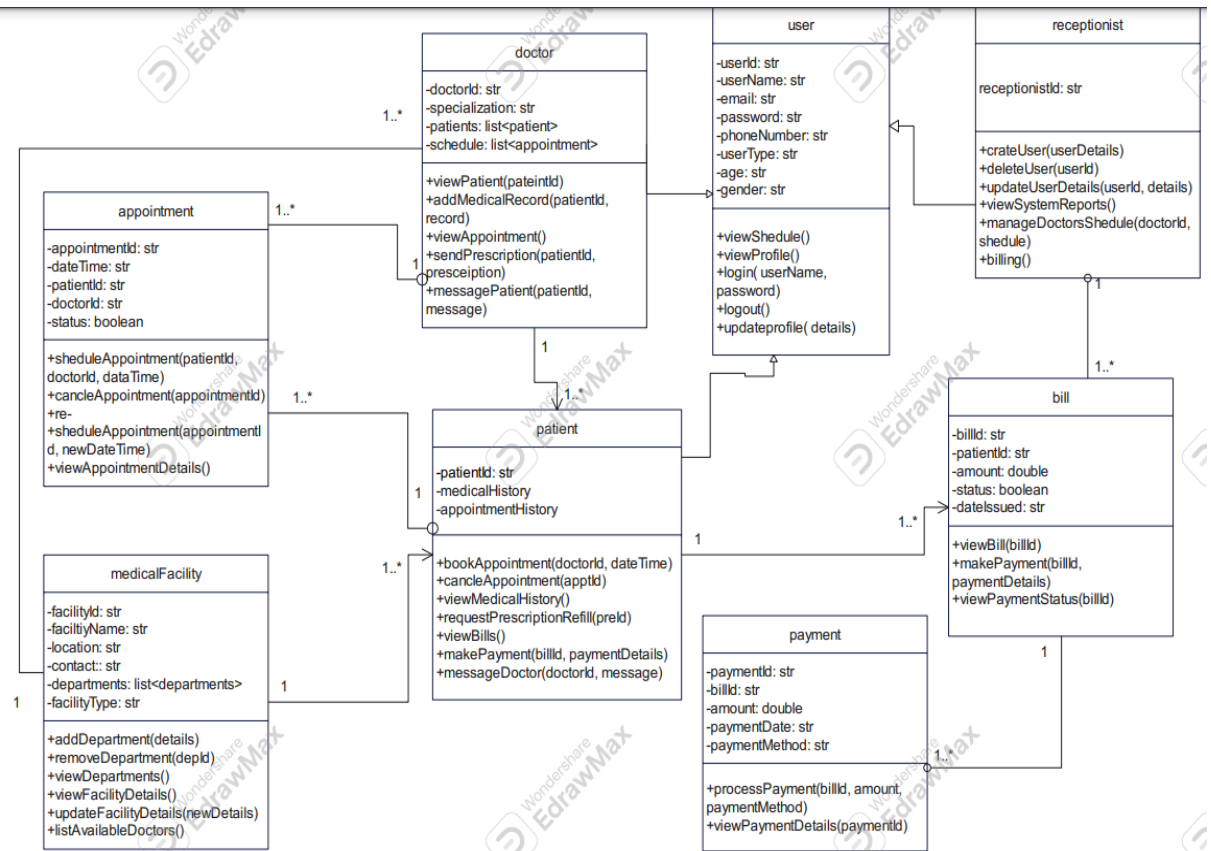
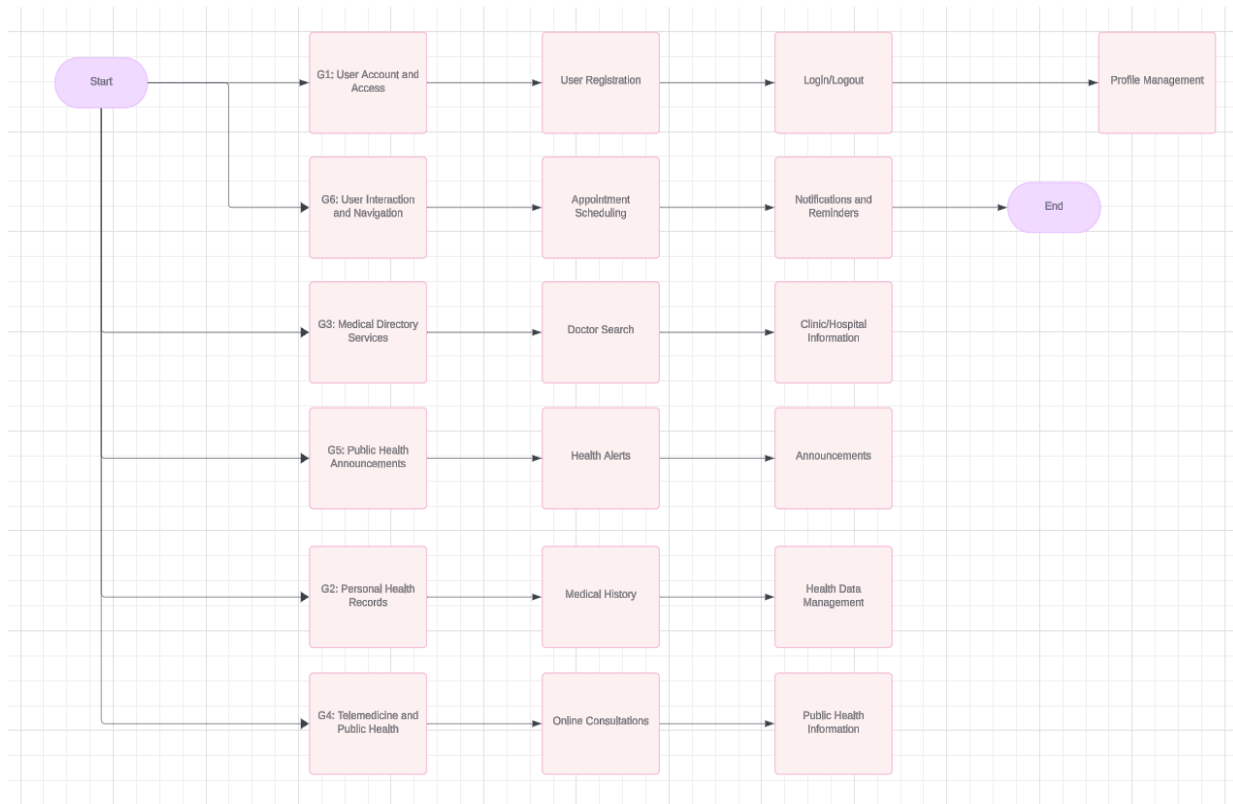


Figure 20: class diagram

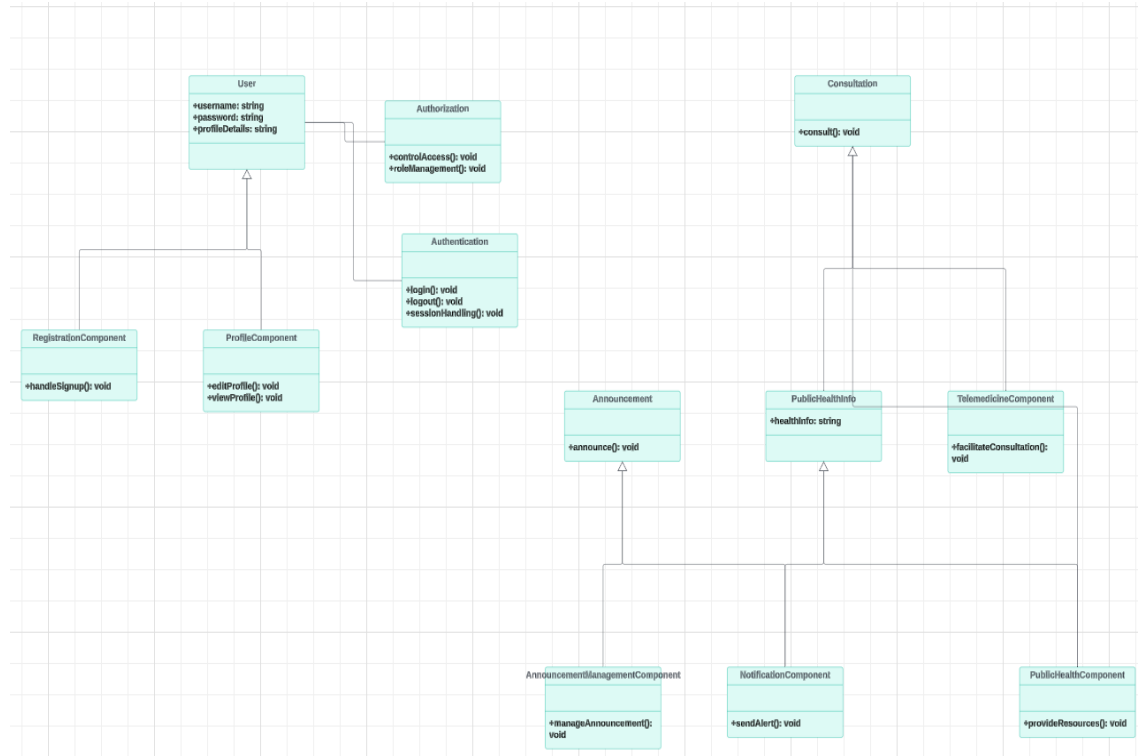
4.4 Architecture Design

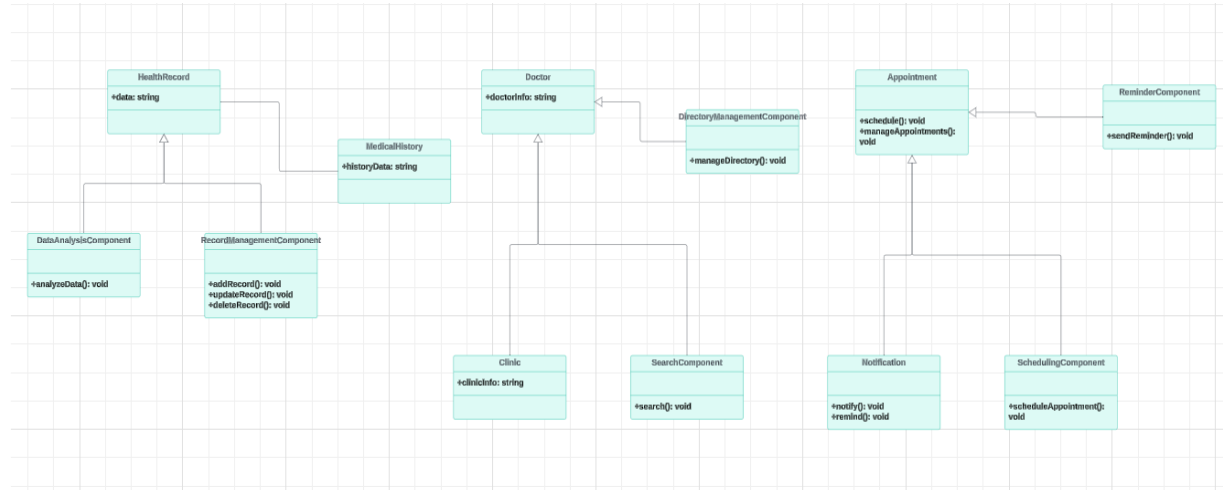


The EasyAppointment system's architecture is divided into modules for account management, health records, service directories, telemedicine, public health information, and user interaction. This structured approach ensures seamless functionality and user-friendly navigation across various health-related services.

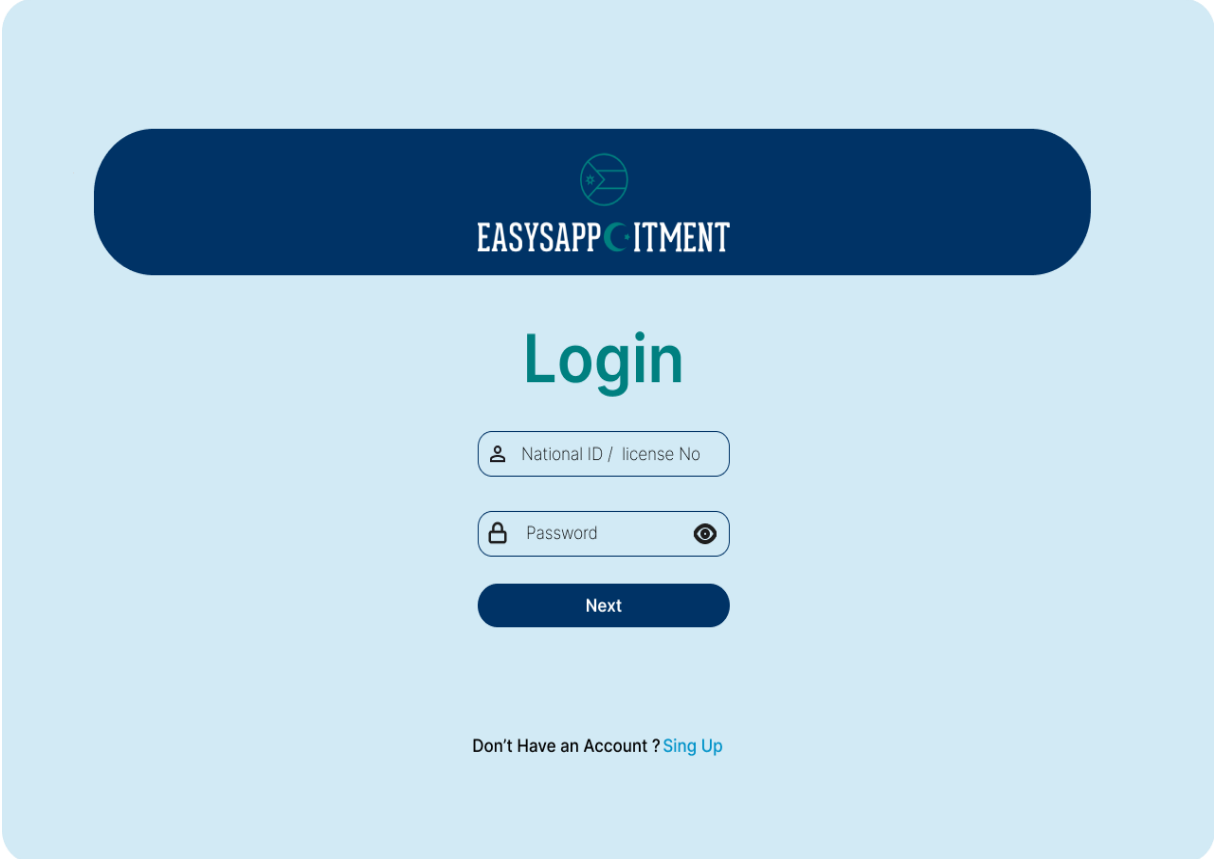
4.5 Classes and Components Design

The following diagrams illustrate the detailed design of the classes and components within the EasyAppointment system. Each module is designed to handle specific functionalities, ensuring a modular and scalable architecture. The diagrams provide a visual representation of the relationships and interactions between different classes and components, facilitating a clear understanding of the system's structure. This modular design enhances the system's flexibility, maintainability, and user-friendliness, providing a robust platform for managing various health-related services.





4.6 Graphical User Interface Design



The login page features a light blue background with a dark blue header bar at the top. The header bar contains the EASYSAPP ITMENT logo, which consists of a circular icon with a stylized 'E' and the text 'EASYSAPP ITMENT' in white. Below the header, the word 'Login' is displayed in a large, bold, teal font. Underneath 'Login', there are two input fields: the first is labeled 'National ID / license No' and the second is labeled 'Password'. The 'Password' field includes a toggle icon (an eye) to the right of the text. Below these fields is a dark blue button with the word 'Next' in white. At the bottom of the page, there is a link that says 'Don't Have an Account ? Sing Up'.

EASYSAPP ITMENT

Login

National ID / license No

Password

Next

Don't Have an Account ? [Sing Up](#)

Figure 21: Login page

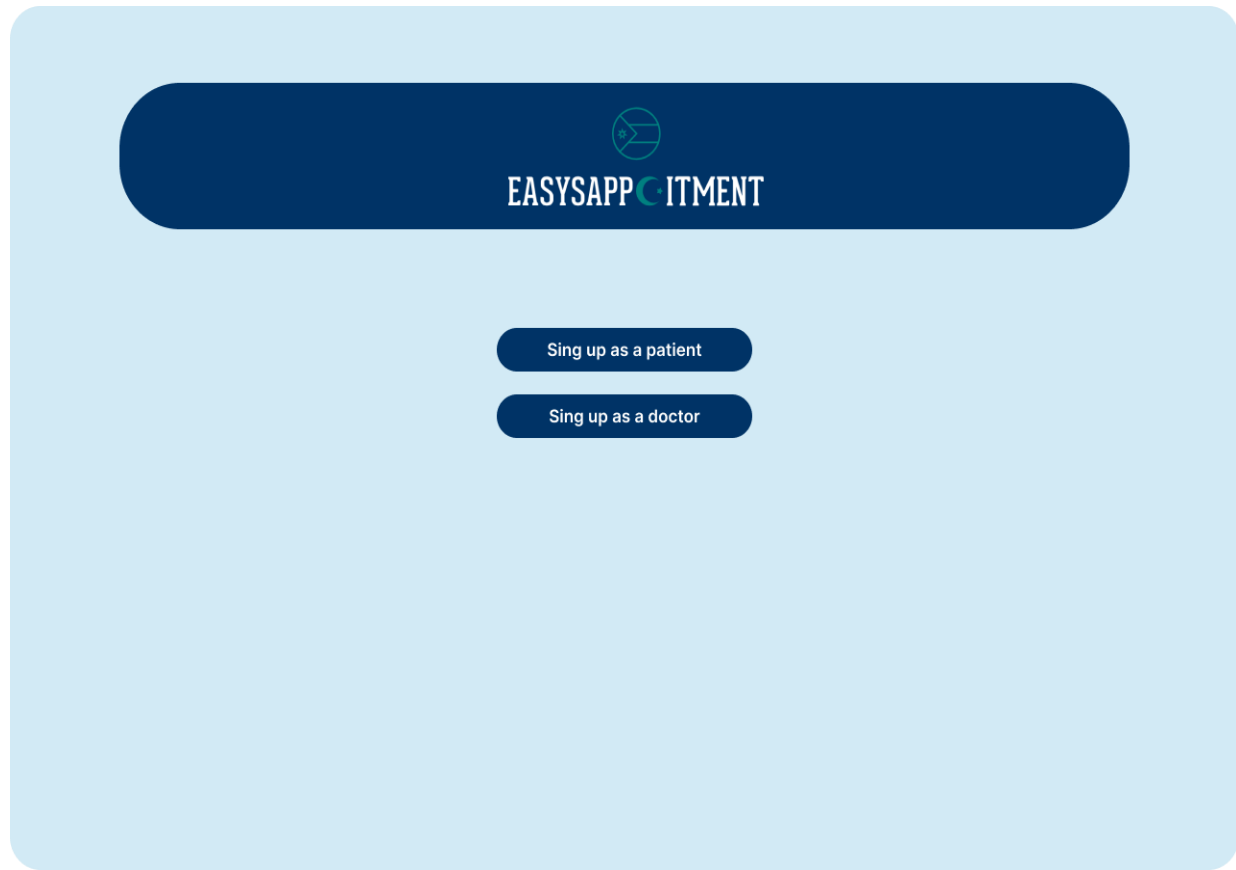


Figure 22: sing Up page

**EASYSAPP ITMENT**

Enter National Number

Enter Phone Number

Enter Email

optional

Enter Password

Register

Figure 23: Sing Up for Patient



EASYSAPP ITMENT

Enter license Number

Enter Phone Number

Enter Email

Enter Password

Register

Figure 24: sing up for Doctors



Figure 25: Home page for Patients

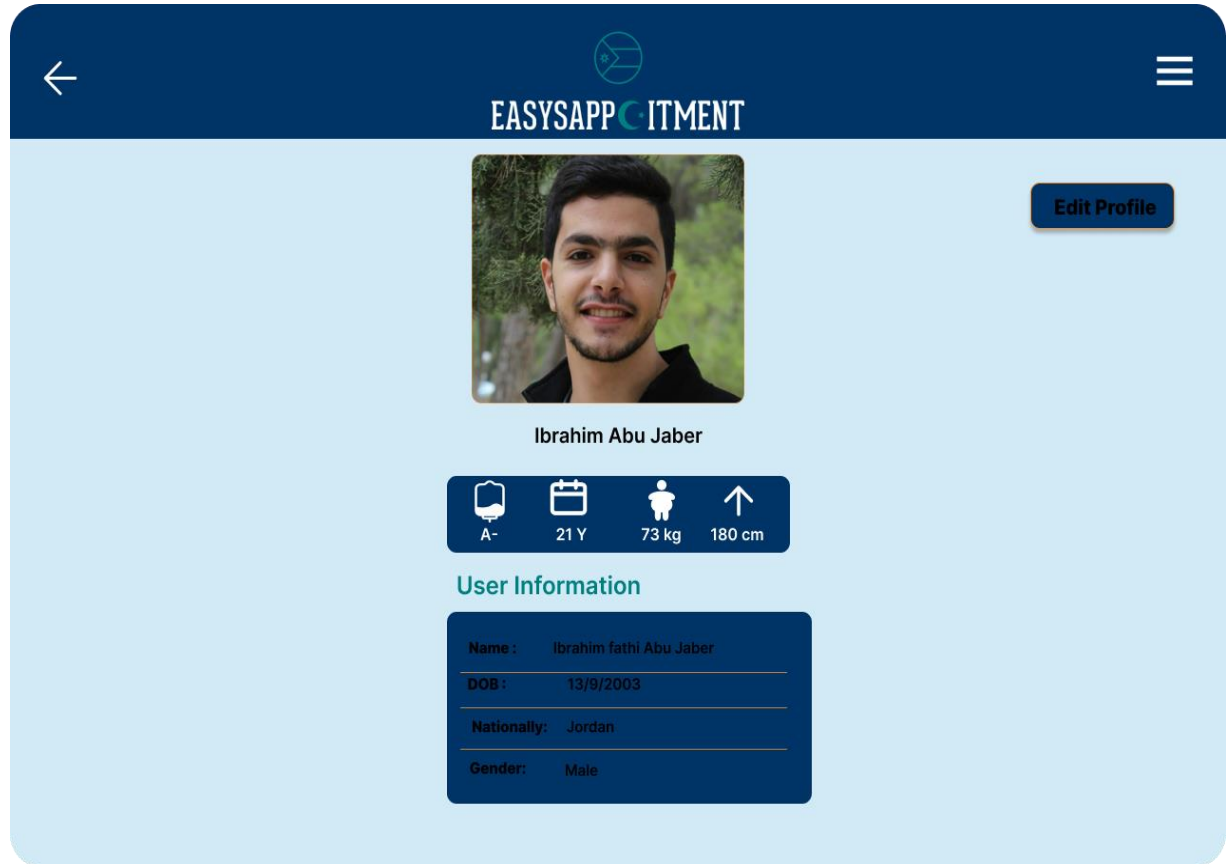


Figure 26: Patient Profile

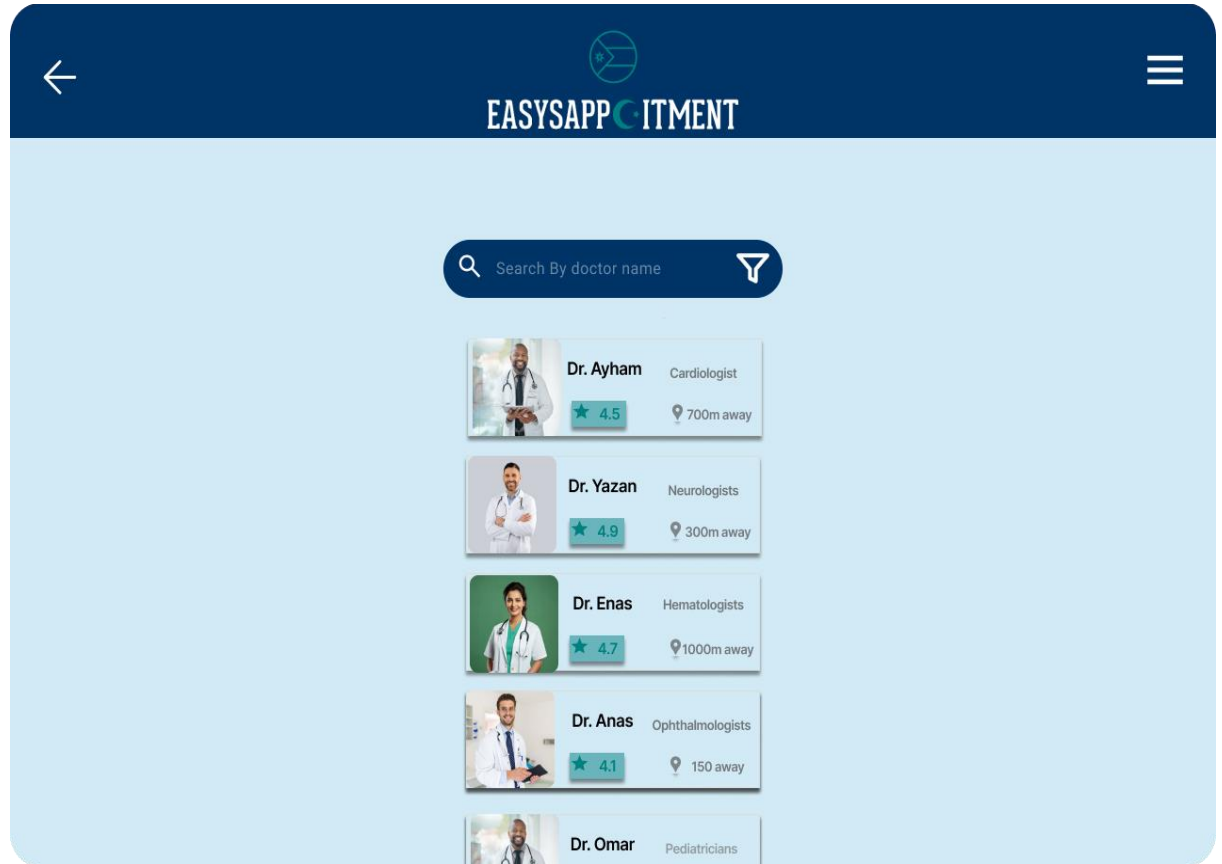


Figure 27: Search for a doctor

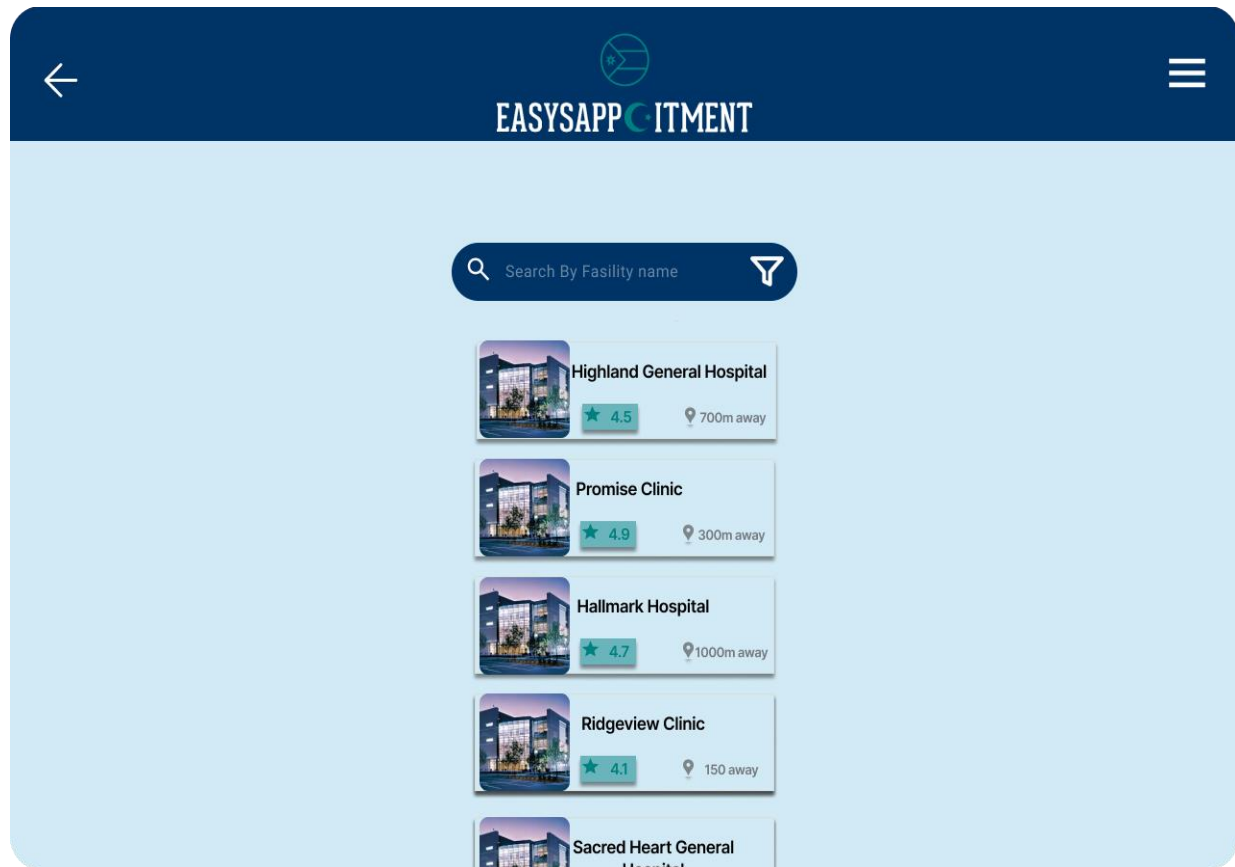


Figure 28: Search for a hospital

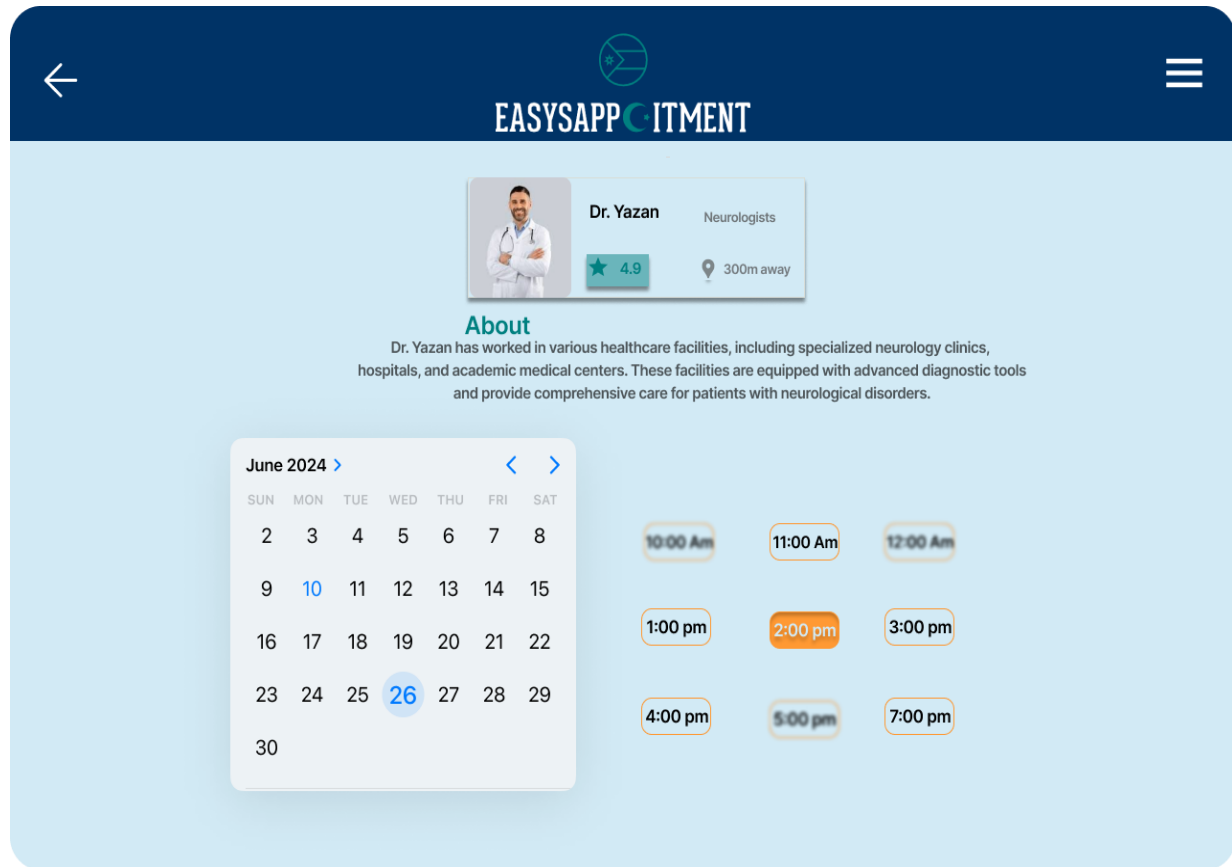


Figure 29: Book an appointment

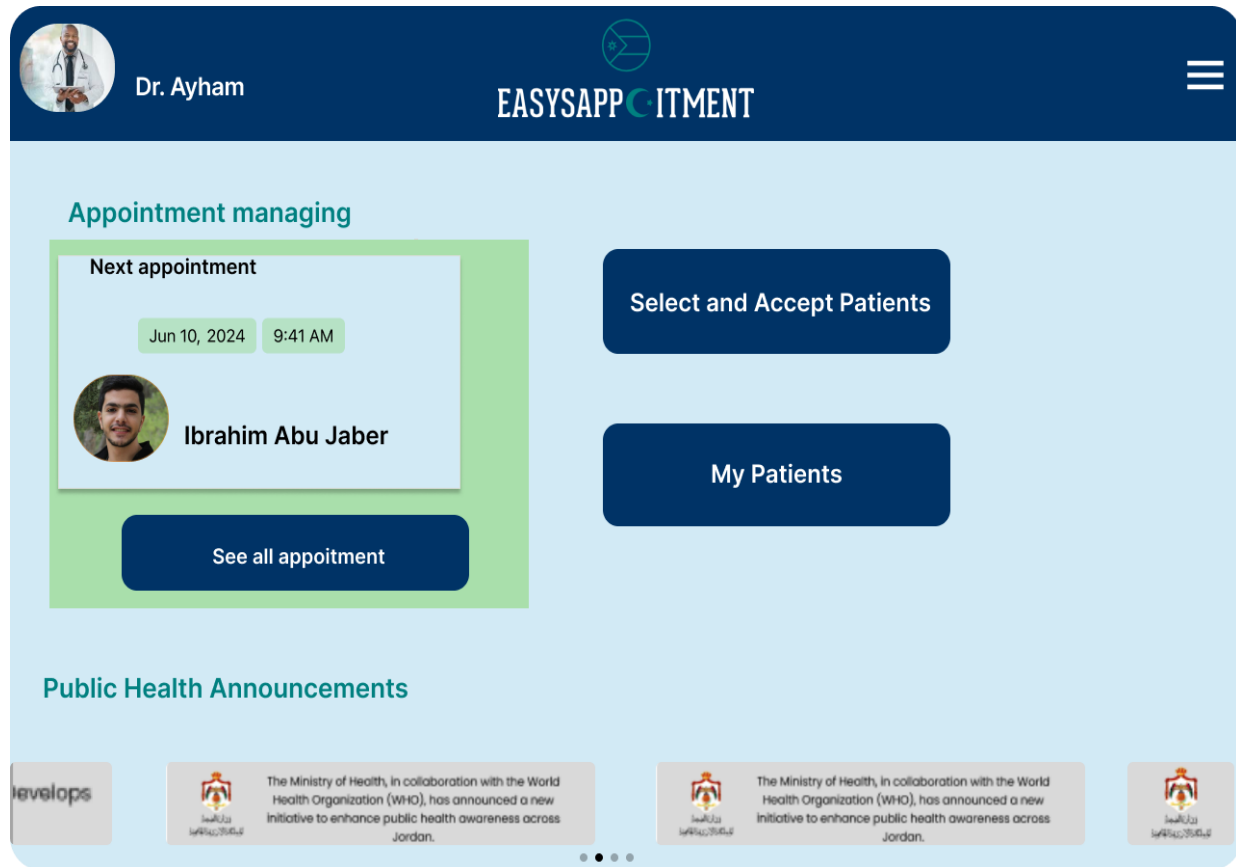


Figure 30: Doctor Home page

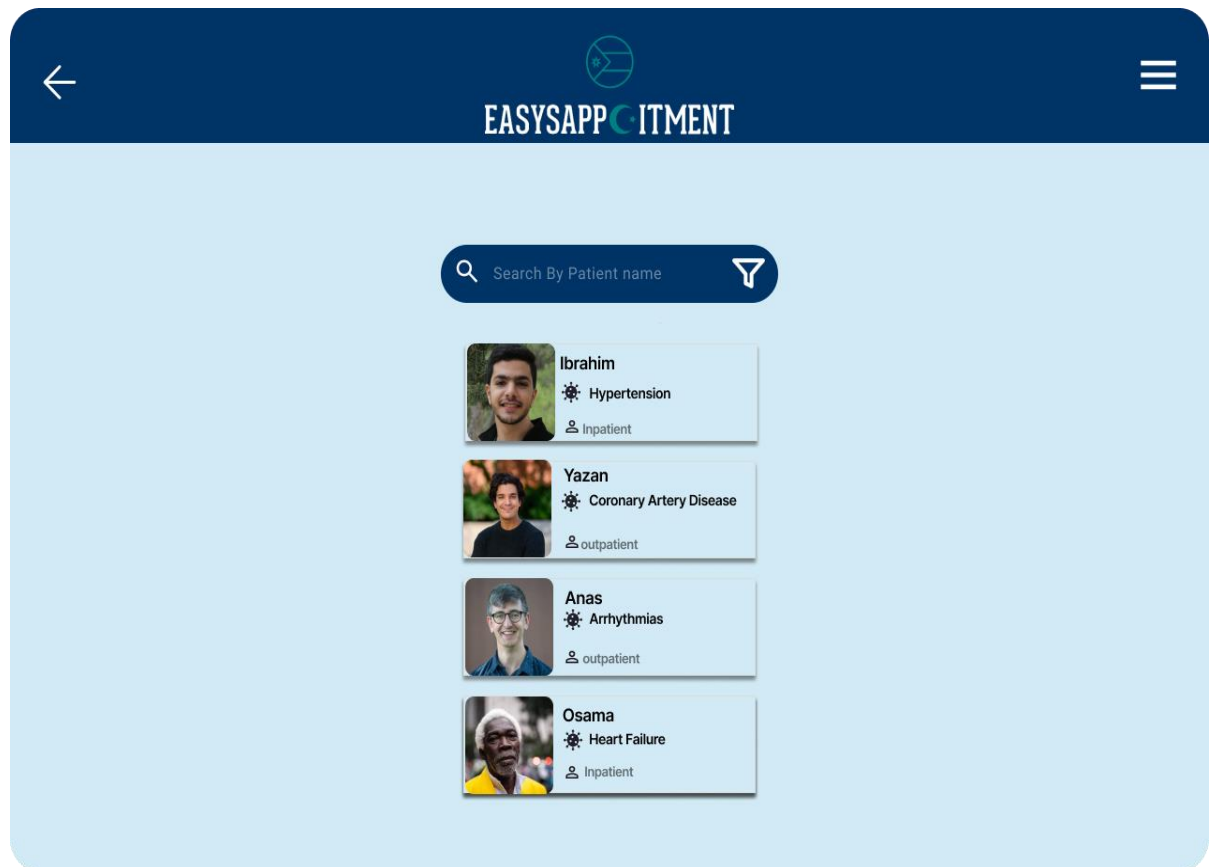


Figure 31: Find Patient

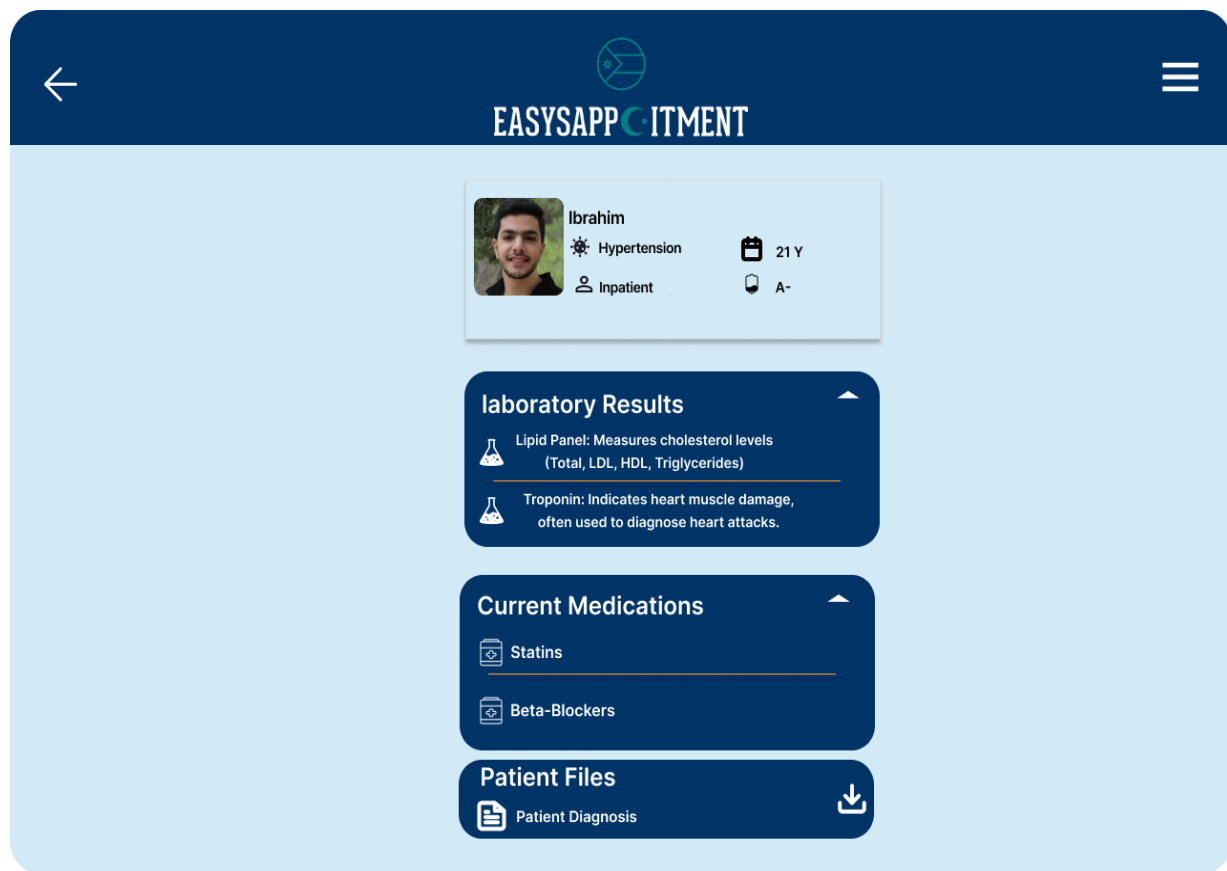


Figure 32: Patient Details

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